SEMICONDUCTOR DESIGN

# CISC · Physical Implementation

## IC Compiler II · StarRC · PrimeTime

Setup & Scenario · Floorplan · Place · CTS · Route · StarRC & PrimeTime · ECO · Filler & Sign-off

DC netlist(cisc, 1.4 ns)를 SAED32 · L-shape boundary에 배치 · 배선하고, 6 corner MCMM과 StarRC · PrimeTime ECO로 sign-off까지.

# Agenda

ICC2 flow 순서대로 진행 — setup부터 StarRC · PrimeTime ECO, sign-off, challenge, 최종 결과까지.

01

### Setup & Scenario

NDM · nlib 구성, routing layer, 1 mode × 6 corner MCMM.

02

### Floorplan & Power Plan

L-shape die, core offset 최소화, ring · mesh · rail.

03

### Place

ss scenario off, half-cycle path group, place_opt.

04

### CTS

CCD clock tree build · route, NDR, post-CTS clock 제약.

05

### Route

pre-route check, route_auto → route_opt, PT 입력 준비.

06

### StarRC & PrimeTime

SPEF 추출, DMSA 4 scenario, fix_eco_timing.

07

### ECO

MUX · U154 clone, U10 LVT, 2차 PT ECO와 route DRC 수정.

08

### Filler & Sign-off

filler 삽입 후 DRC · LVS · PG · 최종 PT 확인.

09

### Challenge

ss scenario까지 켜고 place → CTS → route

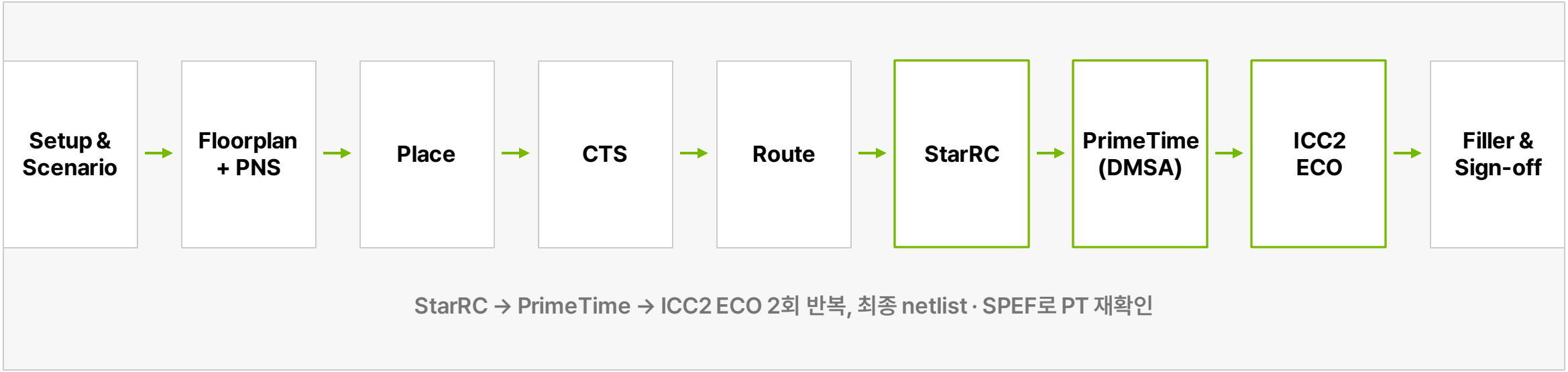
10

### Final Results

단계별 QoR, 최종 sign-off 결과, 총정리.

# Back-end Flow

ICC2로 route까지 진행 → StarRC · PrimeTime으로 ECO → ICC2에 반영 → filler 후 sign-off.




### Scenario

mode normal × corner 6 (tt / ss / ff × 125 °C / −40 °C) = scenario 6개, 각 scenario에 cisc_con.con 적용.

### Main flow

place부터 ss 2 scenario off → 4 scenario로 sign-off. ss까지 켠 run은 09 Challenge.

# CISC Design & Spec

8-bit CISC micro-processor — DESIGN2 (cisc) spec 중 ICC2 관련 항목.

**Function**
8-bit CISC — memory 명령을 fetch → decode → execute, 피보나치 수열 연산 program 수행.

**Hierarchy**
top_mc1: controller / datapath (MDR · IR · ALU · ACC · STATE · PC · ADDR_MUX)
mem1: memory
bid1: bidir_core2
clk_gen1: p_clk · n_clk 생성

**Clock**
p_clk (clk) → top_mc1 / n_clk (clk') → mem1
clk ↔ clk' half-cycle path (0.7 ns)

**I/O**
입력 : clk · reset
출력 : mem_data_in · mem_data_out · mem_data1 · mem_data2 (8-bit)

| 항목 | spec |
|---|---|
| **Logical library** | saed32*vt_{ff1p16vn40c · ff0p85v125c · tt0p78v125c · tt1p05vn40c · ss0p7v125c · ss0p95vn40c}.db |
| **Physical library** | saed32nm_hvt / rvt / lvt_1p9m.lef |
| **Tech** | saed32nm_1p9m_mw.tf |
| **TLU+ / nxtgrd** | saed32nm_1p9m_Cmax / Cmin.tluplus<br>saed32nm_1p9m_Cmax / Cmin.nxtgrd |
| **Map** | saed32nm_tf_itf_tluplus.map |
| **Boundary** | {0 0} {0 84} {56 84} {56 42} {72 42} {72 0}, shape L |
| **Routing layer** | TOP M4 · LOW M1 |
| **Ring / Mesh** | Vertical M4 · Horizontal M3, ring via VIA34SQ_C {2 2} |
| **Design rule** | fanout 40 · cap 5 |
| **Vth** | LVT < 25 % (timing 최적화 시 28 %까지 허용) |

01

SECTION

# Setup & Scenario

logical · physical library로 만든 NDM을 nlib으로 묶고, 6 corner PVT와 scenario 작성.

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# setup.tcl (1/2)

입력 netlist · SDC → corner PVT 정의 → library 생성 → netlist import.

setup.tcl · 0–2

```
##############################################################################
# setup.tcl
#   ICC2 data setup : library -> netlist -> MCMM -> constraints -> PVT -> parasitic -> save
#   Design  : cisc (SAED32nm)
#   Input   : ../dc/output/hier_cisc_netlist.v , ../dc/cisc_con.con
#   Output  : nlib/setup.nlib , reports/setup/*.txt
#   Usage   : icc2_shell -file setup.tcl
##############################################################################

#------------------------------------------------------------------------------
# 0. User variables and output directories
#------------------------------------------------------------------------------
set DESIGN   cisc
set NETLIST  ../dc/output/hier_cisc_netlist.v     ;# DC hierarchical netlist
set SDC_FILE ../dc/cisc_con.con                   ;# DC constraints (applied to every scenario)

# corner : temperature(C) process voltage(V). The name must match create_corner below.
#   *v125c  -> 125C (hot)   *vn40c -> -40C (cold)
set PVT {
    tt0p78v125c   125  1.00  0.78
    tt1p05vn40c   -40  1.00  1.05
    ss0p7v125c    125  0.99  0.70
    ss0p95vn40c   -40  0.99  0.95
    ff0p85v125c   125  1.01  0.85
    ff1p16vn40c   -40  1.01  1.16
}

#------------------------------------------------------------------------------
# 1. Library creation
#    technology lib + reference NDMs (physical-only, HVT / RVT / LVT std cells)
#------------------------------------------------------------------------------
create_lib saed32.nlib \
    -use_technology_lib TECH_LIB.ndm \
    -ref_libs {
        TECH_LIB.ndm
        saed32_physical_only.ndm
        saed32hvt_c.ndm
        saed32rvt_c.ndm
        saed32lvt_c.ndm
    }
save_lib

#------------------------------------------------------------------------------
# 2. Netlist import
#------------------------------------------------------------------------------
read_verilog -top $DESIGN $NETLIST
link_block
```

**Input**

DC hierarchical netlist + cisc_con.con — 모든 scenario에 같은 SDC 적용.

**PVT**

tt / ss / ff × 125 °C / −40 °C = corner 6개, process 1.00 / 0.99 / 1.01.

**nlib**

TECH_LIB.ndm + physical-only · HVT / RVT / LVT NDM을 ref_libs로 묶어 saed32.nlib 생성.

# setup.tcl (2/2)

routing layer → MCMM → SDC → PVT · derate → TLU+.

setup.tcl · 3–7

```
#--------------------------------------------------------------------------------
# 3. Routing layer range : signal routing M1 ~ M4 only
#--------------------------------------------------------------------------------
set_ignored_layers -min_routing_layer M1 -max_routing_layer M4
report_ignored_layers

#--------------------------------------------------------------------------------
# 4. MCMM : 1 mode x 6 corners = 6 scenarios (scenario name = normal_<corner>)
#--------------------------------------------------------------------------------
create_mode normal
foreach {c temp proc volt} $PVT {
    create_corner $c
}
foreach {c temp proc volt} $PVT {
    create_scenario -mode normal -corner $c -name normal_$c
}

#--------------------------------------------------------------------------------
# 5. Constraints : apply the same SDC to every scenario
#--------------------------------------------------------------------------------
foreach {c temp proc volt} $PVT {
    current_scenario normal_$c
    source $SDC_FILE
}

#--------------------------------------------------------------------------------
# 6. Corner PVT and timing derate (per scenario)
#    - temperature / process / VDD voltage from the PVT table
#    - derate: late (data, max) x1.05 , early (min) x0.92 for cell and net delay
#--------------------------------------------------------------------------------
foreach {c temp proc volt} $PVT {
    current_scenario normal_$c
    set_temperature    $temp
    set_process_number $proc
    set_voltage        $volt -object_list VDD
    set_timing_derate -late  1.05 -cell_delay -net_delay
    set_timing_derate -early 0.92 -cell_delay -net_delay
}

#--------------------------------------------------------------------------------
# 7. Parasitic tech (TLU+)
#    - every corner uses the min/max TLU+ pair: early = minTLU , late = maxTLU
#    - extraction temperature follows the corner (125C or -40C)
#--------------------------------------------------------------------------------
foreach {c temp proc volt} $PVT {
    set_parasitic_parameters -corners $c \
        -early_spec minTLU -early_temperature $temp \
        -late_spec  maxTLU -late_temperature  $temp \
        -library TECH_LIB
}
```

**Routing layer**

set_ignored_layers로 signal routing을 M1 – M4로 제한.

**MCMM**

mode normal × corner 6 → scenario 6개 (normal_<corner>).

**Derate**

전 scenario cell · net late 1.05 / early 0.92.

**TLU+**

early minTLU · late maxTLU, 온도는 corner 온도 반영.

# MCMM Scenario

report_scenarios — 1 mode × 6 corner = 6 scenario, setup · hold 모두 active.

report_scenarios

| Name | Mode | Corner | Active | Setup | Hold | Leakage Power | Dynamic Power | Max_tran | Max_cap | Min_cap | Cell EM | Signal EM |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| normal_ff0p85v125c * | | | | | | | | | | | | |
| | normal | ff0p85v125c | true | true | true | true | true | true | true | true | false | false |
| normal_ff1p16vn40c * | | | | | | | | | | | | |
| | normal | ff1p16vn40c | true | true | true | true | true | true | true | true | false | false |
| normal_ss0p7v125c * | | | | | | | | | | | | |
| | normal | ss0p7v125c | true | true | true | true | true | true | true | true | false | false |
| normal_ss0p95vn40c * | | | | | | | | | | | | |
| | normal | ss0p95vn40c | true | true | true | true | true | true | true | true | false | false |
| normal_tt0p78v125c * | | | | | | | | | | | | |
| | normal | tt0p78v125c | true | true | true | true | true | true | true | true | false | false |
| normal_tt1p05vn40c * | | | | | | | | | | | | |
| | normal | tt1p05vn40c | true | true | true | true | true | true | true | true | false | false |

| Corner | tt0p78v125c | tt1p05vn40c | ss0p7v125c | ss0p95vn40c | ff0p85v125c | ff1p16vn40c |
|---|---|---|---|---|---|---|
| **Voltage (V)** | 0.78 | 1.05 | 0.70 | 0.95 | 0.85 | 1.16 |
| **Temp (°C)** | 125 | −40 | 125 | −40 | 125 | −40 |
| **Process** | 1.00 | 1.00 | 0.99 | 0.99 | 1.01 | 1.01 |

**Main flow**

ss 2 scenario는 place부터 off → 4 scenario로 진행.

# 02

SECTION

# Floorplan & Power Plan

L-shape die에 core offset 최소화, M3/M4 ring · mesh와 M1 rail 구성.

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# floorplan.tcl (1/2)

open design → L-shape floorplan → coarse placement → legalize · PG.

floorplan.tcl · 1–4

```
################################################################################
# floorplan.tcl
#   ICC2 floorplan : open design -> floorplan -> coarse placement -> PG -> pins
#                    -> checks -> save
#   Design : cisc (SAED32nm), L-shaped die boundary
#            (bottom 72 x 42 + upper 56 x 42, total height 84)
#   Input  : nlib/setup.nlib , scripts/pns.tcl
#   Output : nlib/floorplan.nlib , TOP.def , reports/floorplan/*.txt
################################################################################

#-------------------------------------------------------------------------------
# 1. Open design
#-------------------------------------------------------------------------------
open_lib   nlib/setup.nlib
open_block /home/AI23/all_lab_student/tools_cisc/icc2/nlib/setup.nlib:cisc.design
link_block

#-------------------------------------------------------------------------------
# 2. Constraints / design rules
#    (none in this script: scenario constraints and PVT were set in setup.tcl)
#-------------------------------------------------------------------------------

#-------------------------------------------------------------------------------
# 3. Floorplan
#    - boundary : L shape, points listed in order
#                 {0 0}->{0 84}->{56 84}->{56 42}->{72 42}->{72 0}
#    - core_offset : one value per boundary edge (6 edges); edge 1 = 3, others = 0
#    - control_type die : the boundary above is the die boundary
#-------------------------------------------------------------------------------
initialize_floorplan \
    -boundary {{0 0} {0 84} {56 84} {56 42} {72 42} {72 0}} \
    -core_offset {3 0 0 0 0 0} \
    -control_type die

# Coarse placement to get a first cell distribution inside the floorplan
create_placement -floorplan
report_utilization

#-------------------------------------------------------------------------------
# 4. Legalize and power / ground
#-------------------------------------------------------------------------------
legalize_placement
connect_pg_net
source scripts/pns.tcl                          ;# PG mesh / ring / std cell rail
```

### Boundary

L-shape die {0 0} {0 84} {56 84} {56 42} {72 42} {72 0}, control_type die.

### Core offset

edge별 값 6개 {3 0 0 0 0 0} — edge 1만 3.

### Coarse placement

create_placement -floorplan → report_utilization.

### PG

legalize_placement → connect_pg_net → pns.tcl.

# floorplan.tcl (2/2)

top-level pin 배치 → PG check · utilization.

floorplan.tcl · 5–7

```
#-------------------------------------------------------------------------------
# 5. Pin placement (top-level ports)
#    - side 1 only (first boundary edge), layer M3, pin spacing 1.5
#-------------------------------------------------------------------------------
set_block_pin_constraints -self -sides {1} -allowed_layers {M3} -pin_spacing_distance 1.5
place_pins -self

#-------------------------------------------------------------------------------
# 6. Checks (printed to screen)
#-------------------------------------------------------------------------------
check_pg_connectivity
check_pg_drc
check_pg_missing_vias
report_utilization

#-------------------------------------------------------------------------------
# 7. Reports to files, save, export
#-------------------------------------------------------------------------------
check_pg_connectivity > reports/floorplan/report_pg_connectivity_after_floorplan.txt
check_pg_drc          > reports/floorplan/report_pg_drc_after_floorplan.txt
check_pg_missing_vias > reports/floorplan/report_pg_missing_vias_after_floorplan.txt
report_utilization    > reports/floorplan/report_utilization_after_floorplan.txt

save_lib -as nlib/floorplan.nlib
write_def -include {rows_tracks cells ports specialnets vias} TOP.def
```

**Pin**

side 1(edge 1)에만 M3 pin, pin spacing 1.5.

**Check / Report**

check_pg_connectivity · check_pg_drc · check_pg_missing_vias · report_utilization.

# Floorplan & Coarse Placement

L-shape die, core offset 최소화 — create_placement -floorplan 결과.

layout · coarse placement



**72 × 84**
Die (μm)
L-shape · 6 edge

**3 / 0**
Core offset (μm)
pin side / 나머지 5 edge

**0.4383**
Utilization
coarse placement

### Boundary

{0 0} → {0 84} → {56 84} → {56 42} → {72 42} → {72 0} — 하단 72 × 42 + 상단 56 × 42.

### Core offset

offset이 작을수록 core 면적 확보 → edge 1(pin side, M3 pin)만 3, 나머지 0으로 최소화.

### Limit

이보다 더 줄이면 route error 발생 → 현재 값이 최소.

# Utilization — Floorplan

report_utilization — coarse placement 시점 0.4383.

report_utilization · after coarse placement

```
****************************************
Report : report_utilization
Design : cisc
Version: T-2022.03-SP5
Date   : Tue Oct 27 17:55:57 2026
****************************************
Utilization Ratio:                      0.4383
Utilization options:
 - Area calculation based on:           site_row of block cisc
 - Categories of objects excluded:      hard_macros macro_keepouts soft_macros io_cells hard_blockages
Total Area:                             5089.2336
Total Capacity Area:                    5089.2336
Total Area of cells:                    2230.6219
Area of excluded objects:
 - hard_macros         :                0.0000
 - macro_keepouts      :                0.0000
 - soft_macros         :                0.0000
 - io_cells            :                0.0000
 - hard_blockages      :                0.0000

Utilization of site-rows with:
 - Site 'unit':                         0.4383

0.4383
```

**0.4383**
**Utilization**
site row 기준

**2,230.62**
**Cell area (μm²)**
DC netlist

**5,089.23**
**Total area (μm²)**
L-shape core

# pns.tcl (1/2)

기존 PG 정리 → ring via master → M3/M4 ring.

pns.tcl · 0–1

```
##############################################################################
# pns.tcl
#   Power network synthesis for cisc (SAED32nm)
#   Order of creation:
#     1) M3/M4 power ring   2) M3/M4 power mesh
#     3) M1 std cell rail + M1->M4 stacked via
#   Called from floorplan.tcl (source scripts/pns.tcl)
##############################################################################

#------------------------------------------------------------------------------
# 0. Clean up previous PG setup and routes
#    - remove all PG strategies / patterns / regions / via rules
#    - remove PG routes created earlier (ring, stripe, macro pin / lib cell pin connections)
#------------------------------------------------------------------------------
remove_pg_strategies -all
remove_pg_patterns -all
remove_pg_regions -all
remove_pg_via_master_rules -all
remove_pg_strategy_via_rules -all
remove_routes -net_types {power ground} -ring -stripe -macro_pin_connect -lib_cell_pin_connect > /dev/null

# Via master used by the ring and mesh intersections (M3-M4, 2x2 cut array)
set_pg_via_master_rule VIA_3_to_4 -contact_code VIA34SQ_C -via_array_dimension {2 2}

#------------------------------------------------------------------------------
# 1. Power ring around the core
#    - M3 horizontal / M4 vertical, width 0.5, spacing 0.3
#    - placed 1.5 outside the core on both directions (offset -1.5 -1.5)
#    - no corner bridge
#------------------------------------------------------------------------------
suppress_message PGR-599

create_pg_ring_pattern P_m3m4_ring \
    -horizontal_layer M3 -horizontal_width {0.5} \
    -vertical_layer   M4 -vertical_width   {0.5} \
    -horizontal_spacing {0.3} -vertical_spacing {0.3} \
    -corner_bridge false \
    -via_rule {{via_master : {VIA_3_to_4}}}

set_pg_strategy S_m3m4_ring -core \
    -pattern { {pattern: P_m3m4_ring} {nets: {VSS VDD}} {offset: -1.5 -1.5}}

# Via rule: connect the ring to existing M3 / M4 straps at their crossings
set_pg_strategy_via_rule S_ring_vias -via_rule { \
    {{{strategies: {S_m3m4_ring}} {layers: {M3}}} {existing: {strap }}{via_master: {VIA_3_to_4}}} \
    {{{strategies: {S_m3m4_ring}} {layers: {M4}}} {existing: {strap }}{via_master: {VIA_3_to_4}}} \
}

compile_pg -strategies {S_m3m4_ring} -via_rule S_ring_vias
```

**Ring via**

VIA_3_to_4 = VIA34SQ_C, array {2 2} — spec ring via와 동일.

**Ring**

M3 (H) / M4 (V), width 0.5 · spacing 0.3, core 바깥 1.5 (offset −1.5).

**Strap via**

ring과 M3 / M4 strap 교차점에 VIA_3_to_4.

# pns.tcl (2/2)

M3/M4 mesh → M1 std cell rail → M1 – M4 stacked via.

pns.tcl · 2–3

```
#-------------------------------------------------------------------------
# 2. Main power mesh
#    - M3 horizontal / M4 vertical, width 0.5, pitch 13.376, VSS/VDD interleaved
#    - M3 offset -2.422 , M4 offset -3 , trimmed at the boundary
#    - via at adjacent-layer intersections (default via master)
#-------------------------------------------------------------------------
create_pg_mesh_pattern P_m3m4 \
    -layers { \
        { {horizontal_layer: M3} {width: 0.5} {spacing: interleaving} {pitch: 13.376} {offset: -2.422} {trim : true} } \
        { {vertical_layer: M4}   {width: 0.5} {spacing: interleaving} {pitch: 13.376} {offset: -3}     {trim : true} } \
    } \
    -via_rule { {intersection: adjacent} {via_master : {default}} }

# Apply the mesh over the core area, start at (10,10),
# extend to the design boundary and create pins there
set_pg_strategy S_vddvss \
    -core \
    -pattern   { {name: P_m3m4} {nets:{VSS VDD}} {offset_start: {10 10}} } \
    -extension { {{stop:design_boundary_and_generate_pin}} }

compile_pg -strategies {S_vddvss}

#-------------------------------------------------------------------------
# 3. Standard cell rail (M1) and M1 -> M4 stacked via
#-------------------------------------------------------------------------
# Per-layer via masters (3x1 cut arrays), stacked to reach M4 from M1
set_pg_via_master_rule pgvia12 -contact_code VIA12SQ -via_array_dimension {3 1}
set_pg_via_master_rule pgvia23 -contact_code VIA23SQ -via_array_dimension {3 1}
set_pg_via_master_rule pgvia34 -contact_code VIA34SQ -via_array_dimension {3 1}

create_pg_std_cell_conn_pattern P_std_cell_rail

# Rails run along every row, extended to the outermost ring on all four sides
set_pg_strategy S_std_cell_rail_VSS_VDD \
    -core \
    -pattern {{pattern: P_std_cell_rail}{nets: {VSS VDD}}} \
    -extension {{stop: outermost_ring}{direction: L B R T }}

# Stack VIA12/23/34 where the M1 rail crosses an existing M4 strap; no vias elsewhere
set_pg_strategy_via_rule S_via_stdcellrail -via_rule { \
    { {{strategies: {S_std_cell_rail_VSS_VDD}}{layers: {M1}}} \
      {{existing: {strap}}{layers: {M4}}} \
      {via_master: {pgvia12 pgvia23 pgvia34}} } \
    { {intersection: undefined}{via_master: NIL} } \
}

compile_pg -strategies {S_std_cell_rail_VSS_VDD} -via_rule S_via_stdcellrail
```

### Mesh

M3 H / M4 V, width 0.5 · pitch 13.376, VSS / VDD interleave, boundary까지 확장 + PG pin 생성.

### Rail

row마다 M1 rail, 4방향 outermost ring까지 확장.

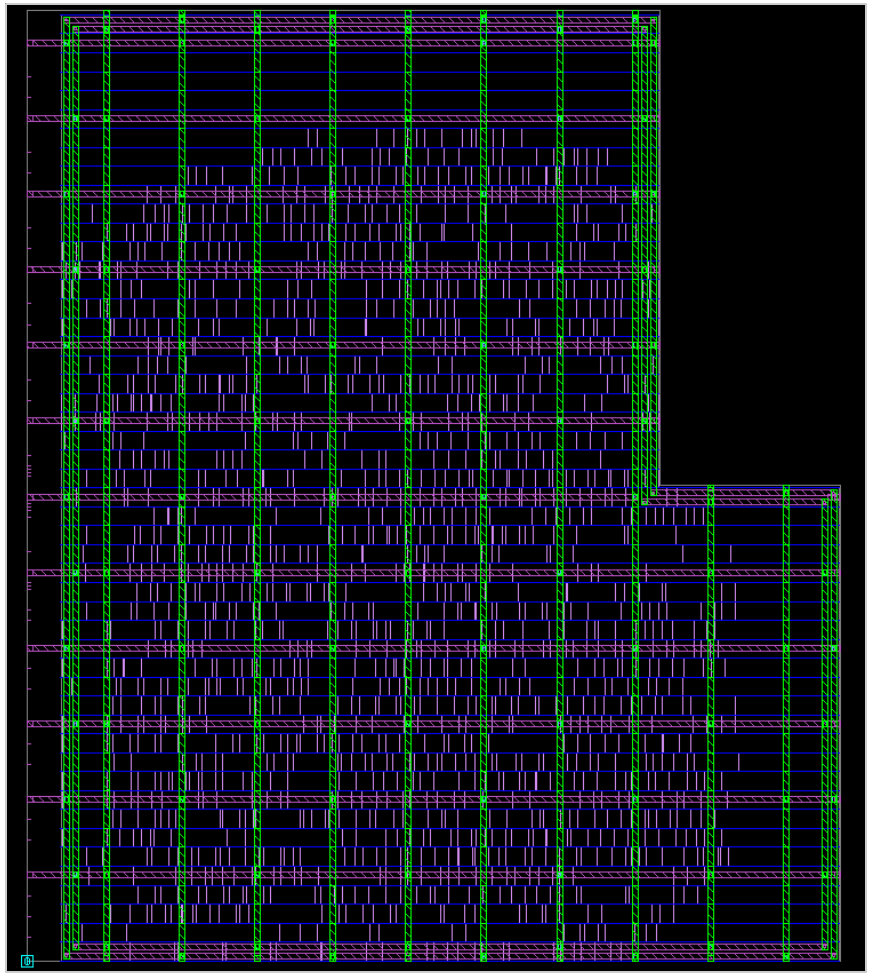
### Stacked via

rail × M4 strap 교차점에만 VIA12 · 23 · 34 (3 × 1).

# Ring & Mesh

L-shape core 둘레 M3/M4 ring, core 위 M3 · M4 mesh, row마다 M1 rail.

layout · after PNS



**Ring**
core를 감싸는 M3 (H) · M4 (V) ring, width 0.5.

**Mesh**
M3 · M4 strap pitch 13.376, boundary까지 확장.

**Rail**
std cell row를 따라 M1 VDD / VSS rail.

# PG Check — Floorplan

check_pg_connectivity · check_pg_drc · check_pg_missing_vias — 위반 없음.

## check_pg_connectivity

```
Number of VDD Wires: 44
Number of VDD Vias: 406
Number of VDD Terminals: 22
**************Verify net VDD connectivity*****************
  Number of floating wires: 0
  Number of floating vias: 0
  Number of floating std cells: 0
  Number of floating hard macros: 0
  Number of floating I/O pads: 0
  Number of floating terminals: 0
  Number of floating hierarchical blocks: 0
************************************************************
Loading cell instances...
Loading P/G wires and vias...
Number of VSS Wires: 42
Number of VSS Vias: 394
Number of VSS Terminals: 22
**************Verify net VSS connectivity*****************
  Number of floating wires: 0
  Number of floating vias: 0
  Number of floating std cells: 0
  Number of floating hard macros: 0
  Number of floating I/O pads: 0
  Number of floating terminals: 0
  Number of floating hierarchical blocks: 0
************************************************************
```

## check_pg_drc · missing_vias

PASS

```
No errors found.
END_CMD: check_pg_drc                CPU:
icc2_shell> check_pg_missing_vias
START_CMD: check_pg_missing_vias CPU
Check net VDD vias...
Number of missing vias: 0
Checking net VDD vias took 0 seconds
Check net VSS vias...
Number of missing vias: 0
Checking net VSS vias took 0 seconds
Overall runtime: 0 seconds.
END_CMD: check_pg_missing_vias CPU:
```

**0**
**Floating**
VDD · VSS 전 항목

**44 / 406**
**VDD wires / vias**
terminal 22

**42 / 394**
**VSS wires / vias**
terminal 22

**0 / 0**
**Missing vias**
VDD / VSS · PG DRC no errors

# 03

SECTION

# Place

ss scenario를 끄고 half-cycle path group을 잡은 뒤 place_opt로 배치 최적화.

정태웅 · 청주대학교

# place.tcl (1/2)

pre-placement check → Vth · fanout 제약 → ss scenario off → congestion 예측.

place.tcl · 0–2

```
##############################################################################
# off_place.tcl
#   ICC2 placement : open floorplan -> Vth/fanout constraints -> scenario setup
#                    -> congestion estimate -> path groups / effort -> place_opt -> save
#   Input  : nlib/floorplan.nlib
#   Output : nlib/off_place.nlib , reports/place/*.txt
##############################################################################

#-----------------------------------------------------------------------------
# 0. Open the floorplan design and check placement readiness
#-----------------------------------------------------------------------------
open_lib   nlib/floorplan.nlib/
open_block /home/AI23/all_lab_student/tools_cisc/icc2/nlib/floorplan.nlib:cisc.design
link_block

check_design -checks pre_placement_stage

#-----------------------------------------------------------------------------
# 1. Optimization constraints
#-----------------------------------------------------------------------------
# Multi-Vth : LVT cells limited to 25% of the design (counted by cell number)
set_threshold_voltage_group_type -type low_vt saed32cell_lvt
set_multi_vth_constraint -low_vt_percentage 25 -cost cell_count

# Max fanout used by optimization
set_app_options -list {opt.common.max_fanout 40}

# Scenarios : the two ss corners are disabled, hold is not optimized in the rest
set_scenario_status normal_ss0p7v125c  -none
set_scenario_status normal_ss0p95vn40c -none
foreach_in_collection scn [all_scenarios] {
    current_scenario $scn
    set_scenario_status $scn -hold false
}

report_scenario

#-----------------------------------------------------------------------------
# 2. Congestion estimate before placement (global route, congestion map only)
#-----------------------------------------------------------------------------
route_global -congestion_map_only true -effort_level low
report_congestion
```

**Multi-Vth**

LVT 25 % (cell_count) — spec LVT < 25 %.

**Fanout**

opt.common.max_fanout 40.

**Scenario**

ss0p7v125c · ss0p95vn40c off(-none), 나머지 4 scenario hold off.

**Congestion**

route_global -congestion_map_only로 배치 전 예측.

# place.tcl (2/2)

high effort + half-cycle path group → place_opt → final_place부터 1회 더.

place.tcl · 3–5

```
#------------------------------------------------------------------------------
# 3. Timing effort and path groups
#    - high effort for timing optimization and for final_place
#    - half-cycle path groups (clk' flops):
#        HC_N2P : clk' flop -> output ports (mem_data_in[*], mem_data2[*])
#        HC_P2N : clk launch -> mem1 *_reg* D pins (clk' flops)
#      weight 2, critical range 0.2
#------------------------------------------------------------------------------
set_app_options -name opt.timing.effort           -value high
set_app_options -name place_opt.final_place.effort -value high

group_path -name HC_N2P -to [get_ports {mem_data_in[*] mem_data2[*]}] -weight 2 -critical_range 0.2
group_path -name HC_P2N -to [get_pins mem1/*_reg*/D]                  -weight 2 -critical_range 0.2
report_path_group

#------------------------------------------------------------------------------
# 4. Placement optimization
#    stages : initial_place -> initial_drc -> initial_opto -> final_place -> final_opto
#------------------------------------------------------------------------------
# 4-1. Full place_opt
place_opt
report_congestion
report_qor -summary -significant_digits 5
report_qor
report_constraints -all_violators -scenarios [all_scenarios]
report_utilization
report_threshold_voltage_groups

# 4-2. Re-run the tail of place_opt (final_place -> final_opto) once more
place_opt -from final_place
report_congestion
report_qor -summary -significant_digits 5
report_qor
report_constraints -all_violators -scenarios [all_scenarios]
report_utilization
report_threshold_voltage_groups

#------------------------------------------------------------------------------
# 5. Reports to files and save
#------------------------------------------------------------------------------
report_threshold_voltage_groups                                > reports/place/report_threshold_voltage_groups_after_off_place.txt
report_congestion                                              > reports/place/report_congestion_after_off_place.txt
report_qor -summary -significant_digits 5                      > reports/place/report_qor_summary_after_off_place.txt
report_qor                                                     > reports/place/report_qor_after_off_place.txt
report_constraints -all_violators -scenarios [all_scenarios]   > reports/place/report_constraints_after_off_place.txt
report_utilization                                             > reports/place/report_utilization_after_off_place.txt

save_lib -as nlib/off_place.nlib
```

**Effort**

opt.timing · place_opt.final_place effort high.

**Path group**

HC_N2P: clk′ flop → mem_data_in · mem_data2 port
HC_P2N: clk launch → mem1 *_reg*/D
weight 2 · critical range 0.2.

**place_opt**

full place_opt 후 -from final_place로 final_place → final_opto 재실행.

# Pre-placement Check

check_design pre_placement_stage — error 0, warning은 모두 무시 가능.

check_design · pre_placement_stage

**0 ERRORS**

```
****************************************
 Report : check_design
 Options: { pre_placement_stage }
 Design : cisc
 Version: T-2022.03-SP5
 Date   : Wed Oct 28 13:22:54 2026
****************************************

Running mega-check 'pre_placement_stage':
    Running atomic-check 'design_mismatch'
    Running atomic-check 'scan_chain'
    Running atomic-check 'mv_design'
    Running atomic-check 'rp_constraints'
    Running atomic-check 'timing'
    Running atomic-check 'hier_pre_placement'

  *** EMS Message summary ***
  ------------------------------------------------------------------------------------------------
  Rule         Type   Count      Message
  ------------------------------------------------------------------------------------------------
  DFT-011      Info   1          The design has no scan chain defined in the scandef.
  TCK-001      Warn   42         The reported endpoint '%endpoint' is unconstrained. Reason: '%re...
  TCK-012      Warn   6          The input port '%port' has no clock_relative delay specified. Mo...
  ------------------------------------------------------------------------------------------------
  Total 49 EMS messages : 0 errors, 48 warnings, 1 info.
  ------------------------------------------------------------------------------------------------

  *** Non-EMS message summary ***
  ------------------------------------------------------------------------------------------------
  Rule         Type   Count      Message
  ------------------------------------------------------------------------------------------------
  NDMUI-173    Info   1          There are no relative placement groups in the design.
  NEX-001      Warn   6          Technology layer '%s' setting '%s' is not valid
  PVT-032      Info   6          Corner %s: no PVT mismatches.
  ------------------------------------------------------------------------------------------------
  Total 13 non-EMS messages : 0 errors, 6 warnings, 7 info.
  ------------------------------------------------------------------------------------------------
```

**TCK-001 (42)**

reset 경로 endpoint — 의도된 unconstrained → 무시.

**TCK-012 (6)**

전부 reset 계열 input port — reset에는 input delay 미적용이 의도 → 무시.

**NEX-001 (6)**

max routing layer를 M4로 잡아 생기는 tech layer 경고 → 무시.

# Congestion — Before Place

report_congestion — overflow GRC 15개 (0.35 %).

report_congestion · before place

```
****************************************
Report : congestion
Design : cisc
Version: T-2022.03-SP5
Date   : Wed Oct 28 13:26:23 2026
****************************************

Layer      |     overflow      |                  # GRCs has
Name       |  total   |   max  | overflow (%)       | max overflow
-------------------------------------------------------------------
Both Dirs  |       15 |      1 |       15  ( 0.35%) |       15
H routing  |       15 |      1 |       15  ( 0.70%) |       15
V routing  |        0 |      0 |        0  ( 0.00%) |        0
```

**15**
**Overflow total**
max 1

**0.35 %**
**GRC overflow**
Both Dirs

**0.70 % / 0.00 %**
**H / V routing**
H 15 GRC · V 0

# place_opt — QoR & Congestion

place_opt 후 tt0p78v125c setup −0.04325 / 23 path — CTS 단계에서 해소.

report_congestion · report_qor -summary

```
****************************************
Report : congestion
Design : cisc
Version: T-2022.03-SP5
Date   : Wed Oct 28 13:30:34 2026
****************************************

Layer     |    overflow     |              # GRCs has
Name      |  total  |  max  | overflow (%)      | max overflow
----------------------------------------------------------------
Both Dirs |       6 |     1 |       6  ( 0.14%) |       6
H routing |       6 |     1 |       6  ( 0.28%) |       6
V routing |       0 |     0 |       0  ( 0.00%) |       0

END_CMD: report_congestion     CPU:    361 s ( 0.10 hr) ELAPSE:    819 s ( 
1
icc2_shell> report_qor -summary -significant_digits 5
****************************************
Report : qor
         -summary
Design : cisc
Version: T-2022.03-SP5
Date   : Wed Oct 28 13:30:34 2026
****************************************

Timing
------------------------------------------------------------------------------
Context                                WNS            TNS            NVE
------------------------------------------------------------------------------
normal_ff0p85v125c (Setup)          0.00848        0.00000              0
normal_ff1p16vn40c (Setup)          0.07045        0.00000              0
normal_tt0p78v125c (Setup)         -0.04325       -0.55868             23
normal_tt1p05vn40c (Setup)          0.03151        0.00000              0
Design             (Setup)         -0.04325       -0.55868             23

Design             (Hold)                --        0.00000              0
------------------------------------------------------------------------------

Miscellaneous
------------------------------------------------------------------------------
Cell Area (netlist):                      2333.04192
Cell Area (netlist and physical only):    2333.04200
Nets with DRC Violations:       2
```

**−0.04325**
Setup WNS (ns)
tt0p78v125c · 23 path

**6 (0.14 %)**
Overflow GRC
Both Dirs · max 1

**2,333.04**
Cell area (μm²)
floorplan 2,230.62

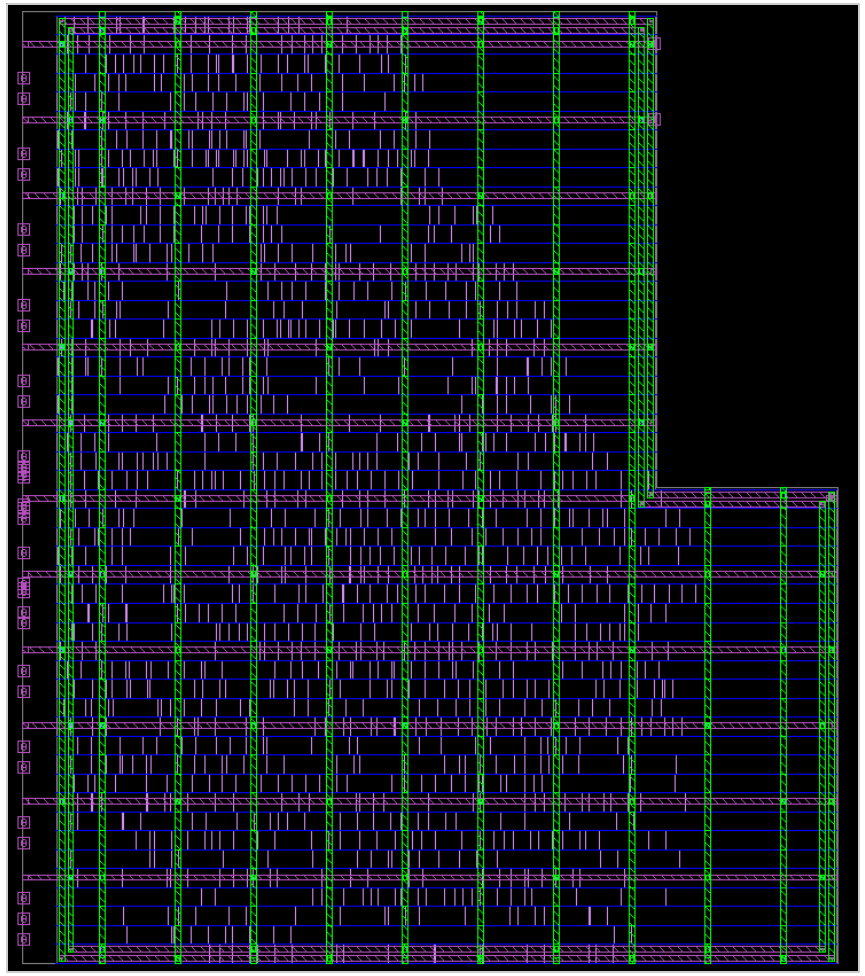
**2**
Nets with DRC
place 시점

**Setup**
ff0p85v125c · ff1p16vn40c · tt1p05vn40c는 met, tt0p78v125c만 위반.

# Place Result

place_opt 완료 layout.

layout · after place



**0.4584**

**Utilization**

cell area / core area

**2,333.04**

**Cell area (μm²)**

floorplan 2,230.62

**6 (0.14 %)**

**Overflow GRC**

Both Dirs · max 1

# 04

SECTION

# CTS

CCD로 clock tree를 build · route하고 propagated clock 기준으로 setup · hold 최적화.

# cts.tcl (1/3)

pre-CTS check → 전역 옵션 → hold scenario → FADD 제외 · 기존 tree 제거.

cts.tcl · 0–2

```
################################################################################
# off_cts.tcl
#   ICC2 clock tree synthesis (CCD enabled) : open placed design -> CTS options
#   -> pre-CTS setup -> clock NDR -> build/route clock -> post-CTS constraints
#   -> final optimization -> save
#   Input  : nlib/off_place.nlib
#   Output : nlib/off_cts.nlib , reports/cts/*.txt
################################################################################

#-------------------------------------------------------------------------------
# 0. Open the placed design and check CTS readiness
#-------------------------------------------------------------------------------
open_lib   nlib/off_place.nlib/
open_block /home/AI23/all_lab_student/tools_cisc/icc2/nlib/off_place.nlib:cisc.design
link_block
check_design -checks pre_clock_tree_stage

#-------------------------------------------------------------------------------
# 1. Global CTS / optimization options
#-------------------------------------------------------------------------------
# Promote clock nets to NDR only where congestion allows
set_app_options -list {cts.optimize.enable_congestion_aware_ndr_promotion true}

# Remove common-path pessimism on reconvergent clock paths (CRPR)
set_app_options -list {time.remove_clock_reconvergence_pessimism true}

# Hold analysis on for all scenarios except the two ss corners
foreach_in_collection scn [all_scenarios] {
    current_scenario $scn
    set_scenario_status $scn -hold true
}
set_scenario_status {normal_ss0p7v125c normal_ss0p95vn40c} -hold false
report_scenarios

#-------------------------------------------------------------------------------
# 2. Pre-CTS setup
#-------------------------------------------------------------------------------
# Disallow full-adder cells (dont_use from synthesis)
set_lib_cell_purpose -include none [get_lib_cells */FADD*]

# Start from a clean clock net (drop any previous tree)
remove_clock_trees -clocks [get_clocks clk]

# Report only : CCD / local skew options
report_app_options *enable_ccd*
report_app_options *local_skew*
```

**Option**

congestion-aware NDR promotion, CRPR on.

**Hold**

ss 2개 제외 4 scenario hold on.

**Pre-CTS**

FADD* cell 사용 제외, 기존 clock tree 제거.

# cts.tcl (2/3)

clock design rule → clock cell → NDR → CCD on → build · route clock.

cts.tcl · 3–5

```
#--------------------------------------------------------------------------------
# 3. CTS design rules
#--------------------------------------------------------------------------------

# Clock net max fanout 40
set_app_options -name cts.common.max_fanout -value 40

# Keep LVT/HVT buffers and inverters out of the clock tree (RVT only)
set_lib_cell_purpose -exclude cts [get_lib_cells "*/*BUF*_LVT */*INV*_LVT */*BUF*_HVT */*INV*_HVT"]

#--------------------------------------------------------------------------------
# 4. Clock NDR
#    - root / internal nets : 2x width / 2x spacing rule on M3-M4
#    - sink nets            : default rule on M1-M4
#--------------------------------------------------------------------------------
create_routing_rule 2xS_2xW_RULE \
    -width    {M1 0.1 M2 0.112 M3 0.112 M4 0.112} \
    -spacings {M1 0.1 M2 0.112 M3 0.112 M4 0.112}

set_clock_routing_rules -clocks [get_clocks clk] -net_type root     -rules 2xS_2xW_RULE -min_routing_layer M3 -max_routing_layer M4
set_clock_routing_rules -clocks [get_clocks clk] -net_type internal -rules 2xS_2xW_RULE -min_routing_layer M3 -max_routing_layer M4
set_clock_routing_rules -clocks [get_clocks clk] -net_type sink     -default_rule       -min_routing_layer M1 -max_routing_layer M4

# Enable CCD (concurrent clock and data optimization) in clock_opt : this flow runs with CCD
set_app_options -name clock_opt.flow.enable_ccd -value true

#--------------------------------------------------------------------------------
# 5. Build and route the clock tree
#--------------------------------------------------------------------------------
clock_opt -from build_clock -to route_clock
```

**Rule**

clock net max_fanout 40.

**Clock cell**

LVT · HVT buffer / inverter 제외 → RVT만.

**NDR**

root · internal 2xS_2xW on M3 – M4, sink default M1 – M4.

**CCD**

clock_opt.flow.enable_ccd true.

# cts.tcl (3/3)

post-CTS clock 제약 → final_opto.

cts.tcl · 6–7

```
#--------------------------------------------------------------------------------
# 6. Post-CTS clock constraints (all scenarios)
#    propagated clock, source latency 0.1,
#    uncertainty setup 0.03 / hold 0.01
#--------------------------------------------------------------------------------
foreach_in_collection scn [all_scenarios] {
    current_scenario $scn
    set_propagated_clock [get_clocks clk]
    set_clock_latency -source 0.1 [get_clocks clk]
    set_clock_uncertainty -setup 0.03 [get_clocks clk]
    set_clock_uncertainty -hold  0.01 [get_clocks clk]
}

#--------------------------------------------------------------------------------
# 7. Final optimization (setup + hold) and reports
#--------------------------------------------------------------------------------
clock_opt -from final_opto
report_qor -summary -significant_digits 5
report_qor
report_constraints -all_violators -scenarios [all_scenarios]
report_clock_qor
report_utilization

report_constraints -all_violators -scenarios [all_scenarios] > reports/cts/report_constraints_after_off_cts.txt
report_qor -summary -significant_digits 5                    > reports/cts/report_qor_summary_after_off_cts.txt
report_qor                                                   > reports/cts/report_qor_after_off_cts.txt
report_clock_qor                                             > reports/cts/report_clock_qor_after_off_cts.txt
report_utilization                                           > reports/cts/report_utilization_after_off_cts.txt

save_lib -as nlib/off_cts.nlib
```

**Propagated**
set_propagated_clock, source latency 0.1.

**Uncertainty**
setup 0.03 / hold 0.01 (skew 제외).

**Final opt**
clock_opt -from final_opto로 setup · hold 최적화.

# Pre-CTS Check

check_design pre_clock_tree_stage — error 0, 앞 단계에 없던 warning만 설명.

check_design · pre_clock_tree_stage

**0 ERRORS**

```
****************************************
 Report : check_design
 Options: { pre_clock_tree_stage }
 Design : cisc
 Version: T-2022.03-SP5
 Date   : Wed Oct 28 14:00:58 2026
****************************************

Running mega-check 'pre_clock_tree_stage':
    Running atomic-check 'design_mismatch'
    Running atomic-check 'scan_chain'
    Running atomic-check 'mv_design'
    Running atomic-check 'legality'
    Running atomic-check 'timing'
    Running atomic-check 'clock_trees'
    Running atomic-check 'hier_pre_clock_tree'

  *** EMS Message summary ***
  ------------------------------------------------------------------------------------------------------
  Rule         Type   Count      Message
  ------------------------------------------------------------------------------------------------------
  DFT-011      Info   1          The design has no scan chain defined in the scandef.
  TCK-001      Warn   42         The reported endpoint '%endpoint' is unconstrained. Reason: '%re...
  TCK-012      Warn   6          The input port '%port' has no clock_relative delay specified. Mo...
  ------------------------------------------------------------------------------------------------------
  Total 49 EMS messages : 0 errors, 48 warnings, 1 info.
  ------------------------------------------------------------------------------------------------------

  *** Non-EMS message summary ***
  ------------------------------------------------------------------------------------------------------
  Rule         Type   Count      Message
  ------------------------------------------------------------------------------------------------------
  CTS-101      Info   1          %s will work on the following scenarios.
  CTS-107      Info   1          %s will work on all clocks in active scenarios, including %d mas...
  CTS-973      Info   1          The value of option cts.compile.enable_cell_relocation has been ...
  NEX-001      Warn   6          Technology layer '%s' setting '%s' is not valid
  PDC-003      Warn   14         Routing direction of metal layer %s is neither "horizontal" nor ...
  PVT-032      Info   6          Corner %s: no PVT mismatches.
  ------------------------------------------------------------------------------------------------------
  Total 29 non-EMS messages : 0 errors, 20 warnings, 9 info.
  ------------------------------------------------------------------------------------------------------
```

**PDC-003 (14)**

max routing layer M4 설정으로 그 위 layer에 routing direction 없음 → 무시.

# QoR — After CTS

propagated clock 기준 4 scenario setup · hold 위반 0 — place 잔여 setup 해소.

report_qor -summary · after CTS

**0 NVE**

```
****************************************
Report : qor
         -summary
Design : cisc
Version: T-2022.03-SP5
Date   : Wed Oct 28 14:07:48 2026
****************************************
Information: Timer using 'CRPR'. (TIM-050)

Timing
--------------------------------------------------------------------------------
Context                                 WNS             TNS             NVE
--------------------------------------------------------------------------------
normal_ff0p85v125c (Setup)          0.02505         0.00000               0
normal_ff1p16vn40c (Setup)          0.09029         0.00000               0
normal_tt0p78v125c (Setup)          0.00017         0.00000               0
normal_tt1p05vn40c (Setup)          0.04441         0.00000               0
Design             (Setup)          0.00017         0.00000               0

normal_ff0p85v125c (Hold)           0.08145         0.00000               0
normal_ff1p16vn40c (Hold)           0.03779         0.00000               0
normal_tt0p78v125c (Hold)           0.13576         0.00000               0
normal_tt1p05vn40c (Hold)           0.07806         0.00000               0
Design             (Hold)           0.03779         0.00000               0
--------------------------------------------------------------------------------

Miscellaneous
--------------------------------------------------------------------------------
Cell Area (netlist):                           2242.05837
Cell Area (netlist and physical only):         2242.05800
Nets with DRC Violations:       0
```

**+0.00017**
Setup WNS (ns)
tt0p78v125c

**+0.03779**
Hold WNS (ns)
ff1p16vn40c

**2,242.06**
Cell area (μm²)
place 2,333.04

**0**
Nets with DRC

# Clock QoR — After CTS

clk 90 sink · 2 level · repeater 2, trans · cap DRC 0.

## report_clock_qor · after CTS

```
============================================================ Summary Table for Corner ff0p85v125c ============================================================
Clock /                                 Attrs     Sinks Levels    Clock     Clock     Clock       Max    Global  Trans DRC  Cap DRC      Wire
Skew Group                                                       Repeater  Repeater   Stdcell   Latency     Skew     Count     Count    Length
                                                                   Count      Area      Area
------------------------------------------------------------------------------------------------------------------------------------------------
### Mode: normal, Scenario: normal_ff0p85v125c
clk                                      M,D        90      2        2      6.35      6.35      0.06      0.06         0         0    463.42
------------------------------------------------------------------------------------------------------------------------------------------------
All Clocks                                          90      2        2      6.35      6.35      0.06      0.06         0         0    463.42


Warning: Please use -largest / -smallest / -all switches with -show_verbose_paths / -show_paths to report the clock paths. (CTS-956)
================================================
==== Summary Reporting for Corner ff1p16vn40c ====
================================================

============================================================ Summary Table for Corner ff1p16vn40c ============================================================
Clock /                                 Attrs     Sinks Levels    Clock     Clock     Clock       Max    Global  Trans DRC  Cap DRC      Wire
Skew Group                                                       Repeater  Repeater   Stdcell   Latency     Skew     Count     Count    Length
                                                                   Count      Area      Area
------------------------------------------------------------------------------------------------------------------------------------------------
### Mode: normal, Scenario: normal_ff1p16vn40c
clk                                      M,D        90      2        2      6.35      6.35      0.04      0.04         0         0    463.42
------------------------------------------------------------------------------------------------------------------------------------------------
All Clocks                                          90      2        2      6.35      6.35      0.04      0.04         0         0    463.42


Warning: Please use -largest / -smallest / -all switches with -show_verbose_paths / -show_paths to report the clock paths. (CTS-956)
================================================
==== Summary Reporting for Corner tt0p78v125c ====
================================================

============================================================ Summary Table for Corner tt0p78v125c ============================================================
Clock /                                 Attrs     Sinks Levels    Clock     Clock     Clock       Max    Global  Trans DRC  Cap DRC      Wire
Skew Group                                                       Repeater  Repeater   Stdcell   Latency     Skew     Count     Count    Length
                                                                   Count      Area      Area
------------------------------------------------------------------------------------------------------------------------------------------------
### Mode: normal, Scenario: normal_tt0p78v125c
clk                                      M,D        90      2        2      6.35      6.35      0.11      0.11         0         0    463.42
------------------------------------------------------------------------------------------------------------------------------------------------
All Clocks                                          90      2        2      6.35      6.35      0.11      0.11         0         0    463.42


Warning: Please use -largest / -smallest / -all switches with -show_verbose_paths / -show_paths to report the clock paths. (CTS-956)
================================================
==== Summary Reporting for Corner tt1p05vn40c ====
================================================

============================================================ Summary Table for Corner tt1p05vn40c ============================================================
Clock /                                 Attrs     Sinks Levels    Clock     Clock     Clock       Max    Global  Trans DRC  Cap DRC      Wire
Skew Group                                                       Repeater  Repeater   Stdcell   Latency     Skew     Count     Count    Length
                                                                   Count      Area      Area
------------------------------------------------------------------------------------------------------------------------------------------------
### Mode: normal, Scenario: normal_tt1p05vn40c
clk                                      M,D        90      2        2      6.35      6.35      0.06      0.06         0         0    463.42
------------------------------------------------------------------------------------------------------------------------------------------------
All Clocks                                          90      2        2      6.35      6.35      0.06      0.06         0         0    463.42
```

- **0.11 / 0.11**
  **tt0p78v125c**
  latency / skew (ns)
- **0.06 / 0.06**
  **tt1p05vn40c**
  latency / skew (ns)
- **0.06 / 0.06**
  **ff0p85v125c**
  latency / skew (ns)
- **0.04 / 0.04**
  **ff1p16vn40c**
  latency / skew (ns)

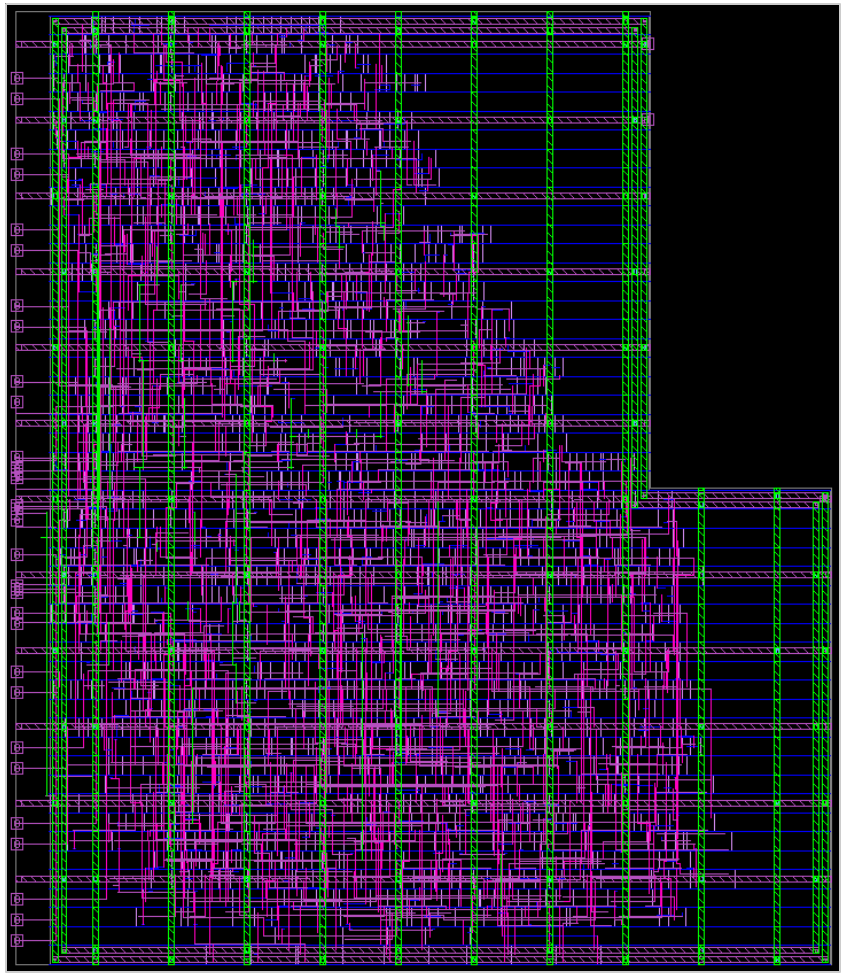
### Skew

CCD on → half-cycle path (clk → clk') setup 확보를 위해 capture clock을 늦춤 (useful skew).

# CTS Result

clock tree 배선 완료 layout.

layout · after CTS



**90 / 2**

**Sinks / Levels**

clk

**2 · 6.35 µm²**

**Clock repeaters**

RVT only

**0 / 0**

**Trans / Cap DRC**

4 scenario

**0.4406**

**Utilization**

cell area 2,242.06

# 05

SECTION

# Route

routability 점검 후 route_auto → route_opt, PrimeTime 입력 준비까지.

정태웅 · 청주대학교

# route.tcl (1/3)

pre-route check → redundant via → route 옵션.

route.tcl · 0–2

```
##############################################################################
# off_route.tcl
#   ICC2 routing : pre-route checks -> route_auto -> route_opt -> post-route checks
#                  -> save -> export for PrimeTime
#   Input  : nlib/off_cts.nlib
#   Output : nlib/off_route.nlib ,
#            outputs_icc2/pt_off_cisc_icc.v , outputs_icc2/cisc_<scenario>.sdc
##############################################################################

#------------------------------------------------------------------------------
# 0. Open the CTS design and run pre-route checks
#------------------------------------------------------------------------------
open_lib   nlib/off_cts.nlib/
open_block /home/AI23/all_lab_student/tools_cisc/icc2/nlib/off_cts.nlib:cisc.design
link_block

check_routability
check_design -checks pre_route_stage

#------------------------------------------------------------------------------
# 1. Redundant via insertion (after detail route)
#------------------------------------------------------------------------------
# medium : default mapping, 2-cut redundant via
set_app_options -name route.common.post_detail_route_redundant_via_insertion -value medium
# list the via mapping only (does not insert vias)
add_redundant_vias -list_only true

#------------------------------------------------------------------------------
# 2. Route options : timing-driven, crosstalk-driven only in track assignment
#------------------------------------------------------------------------------
set_app_options -list {
    route.global.timing_driven     true
    route.global.crosstalk_driven  false
    route.track.timing_driven      true
    route.track.crosstalk_driven   true
    route.detail.timing_driven     true
}
```

**Pre-route**

check_routability · check_design pre_route_stage.

**Redundant via**

post detail route redundant via insertion medium.

**Option**

global · track · detail timing-driven, crosstalk-driven은 track만.

# route.tcl (2/3)

route_auto → SI on → route_opt 2회 → post-route check.

route.tcl · 3–5

```
#-------------------------------------------------------------------------------
# 3. Global route -> track assignment -> detail route
#-------------------------------------------------------------------------------
route_auto
check_routes

#-------------------------------------------------------------------------------
# 4. Post-route optimization with SI analysis
#-------------------------------------------------------------------------------
set_app_options -list {time.si_enable_analysis true}

route_opt
report_qor -summary -significant_digits 5

route_opt
report_qor -summary -significant_digits 5

#-------------------------------------------------------------------------------
# 5. Post-route checks
#-------------------------------------------------------------------------------
check_routes
check_legality
check_lvs -open_reporting detailed -check_child_cells true -max_errors 9999
check_pg_drc
check_pg_connectivity

#-------------------------------------------------------------------------------
# 6. Save
#-------------------------------------------------------------------------------
save_lib -as nlib/off_route.nlib
```

**route_auto**
global route → track assignment → detail route.

**route_opt**
time.si_enable_analysis true 후 route_opt 2회.

**Check**
check_routes · check_legality · check_lvs · check_pg_drc · check_pg_connectivity.

# route.tcl (3/3)

PrimeTime 입력 준비 — netlist와 scenario별 SDC.

route.tcl · 7

```
#-------------------------------------------------------------------------
# 7. Export for PrimeTime (renamed netlist + per-scenario SDC)
#-------------------------------------------------------------------------
save_block -as cisc_pt
open_block /home/AI23/all_lab_student/tools_cisc/icc2/nlib/off_route.nlib:cisc_pt.design
current_block cisc_pt
link_block

# Verilog-legal names, hierarchical; physical-only cells skipped; changes logged
change_names -rules verilog -hierarchy -skip_physical_only_cells -log_changes cisc_names.log
write_verilog -exclude {all_physical_cells} outputs_icc2/pt_off_cisc_icc.v

# SDC for the 4 scenarios analyzed in PrimeTime (ss corners excluded)
foreach sc {normal_ff0p85v125c normal_ff1p16vn40c normal_tt0p78v125c normal_tt1p05vn40c} {
    write_sdc -output outputs_icc2/cisc_${sc}.sdc -scenario $sc -nosplit
}
```

**Netlist**

change_names -rules verilog, physical-only cell 제외.

**SDC**

ss 제외 4 scenario (ff0p85v125c · ff1p16vn40c · tt0p78v125c · tt1p05vn40c).

# Pre-route Check

check_design pre_route_stage — 앞 단계에 없던 항목, 모두 영향 없음.

check_design · pre_route_stage **CR-030 7**

```
****************************************
 Report : check_design
 Options: { pre_route_stage }
 Design : cisc
 Version: T-2022.03-SP5
 Date   : Wed Oct 28 18:20:03 2026
****************************************

Running mega-check 'pre_route_stage':
    Running atomic-check 'design_mismatch'
    Running atomic-check 'scan_chain'
    Running atomic-check 'mv_design'
    Running atomic-check 'timing'
    Running atomic-check 'routability'
    Running atomic-check 'hier_pre_route'

  *** EMS Message summary ***
  ------------------------------------------------------------------------------------------------------------
  Rule         Type    Count      Message
  ------------------------------------------------------------------------------------------------------------
  CR-030       Error   7          Port %instance %port at (%x1 %y1)_(%x2 %y2) on %layers is out of...
  DFT-011      Info    1          The design has no scan chain defined in the scandef.
  TCK-001      Warn    42         The reported endpoint '%endpoint' is unconstrained. Reason: '%re...
  TCK-012      Warn    6          The input port '%port' has no clock_relative delay specified. Mo...
  ------------------------------------------------------------------------------------------------------------
  Total 56 EMS messages : 7 errors, 48 warnings, 1 info.
  ------------------------------------------------------------------------------------------------------------

  *** Non-EMS message summary ***
  ------------------------------------------------------------------------------------------------------------
  Rule         Type    Count      Message
  ------------------------------------------------------------------------------------------------------------
  NEX-001      Warn    6          Technology layer '%s' setting '%s' is not valid
  PVT-032      Info    6          Corner %s: no PVT mismatches.
  ZRT-022      Warn    1          Cannot find a default contact code for layer %s.
  ZRT-025      Warn    6          Layer %s does not have a preferred direction, assigning to %s.
  ZRT-619      Info    1          Via ladder engine would be activated for pattern must join conne...
  ZRT-703      Info    1          Option route.detail.force_end_on_preferred_grid will be ignored ...
  ZRT-706      Info    1          When applicable layer based tapering will taper up to %.2f in di...
  ZRT-707      Info    1          When applicable Min-max layer allow_pin_connection mode will all...
  ZRT-718      Info    1          When applicable Min-max layer allow_pin_connection mode will all...
  ZRT-720      Warn    7          Port  %s at {{%.3f,%.3f} {%.3f,%.3f}} is out of boundary. Cannot...
  ZRT-729      Warn    744        Track at {%.3f %.3f} {%.3f %.3f} on layer %s is out of boundary.
  ------------------------------------------------------------------------------------------------------------
  Total 775 non-EMS messages : 0 errors, 764 warnings, 11 info.
  ------------------------------------------------------------------------------------------------------------
```

**CR-030 (7) · ZRT-720 (7)**
LVS short / open 0, PG floating 0 → 영향 없음.

**ZRT-729 (744)**
boundary 밖 track — route 후 DRC 0 → 영향 없음.

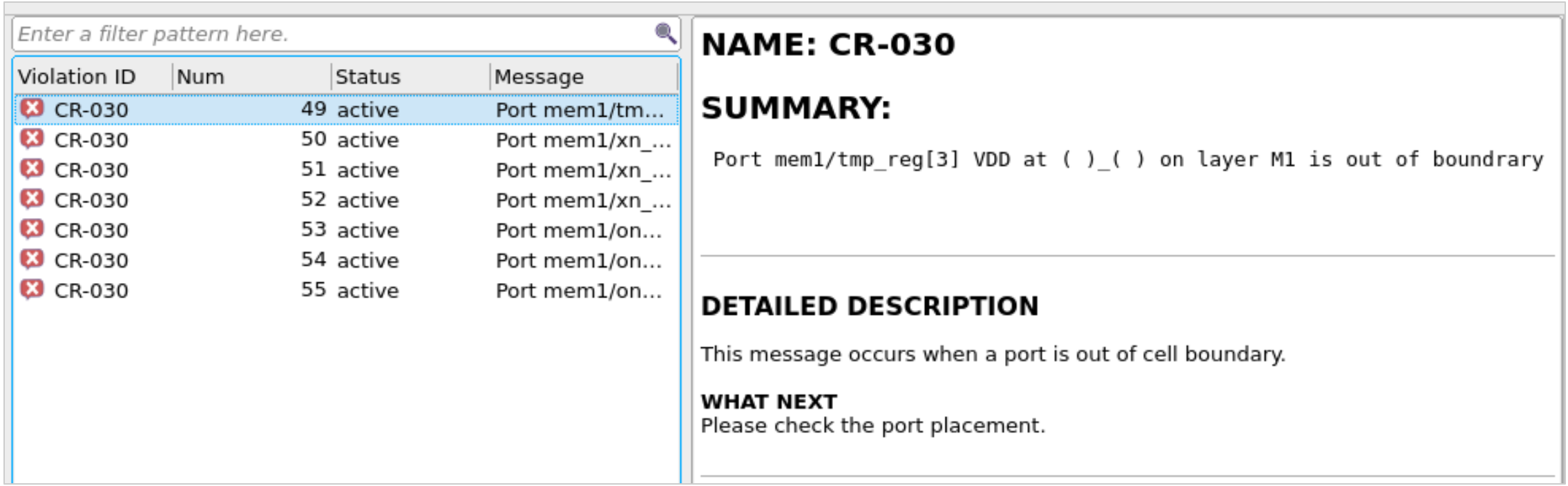
**ZRT-025 (6)**
max routing layer M4 위 layer의 direction 자동 지정 → 무시.

**ZRT-022 (1)**
non-routing layer → 무시.

# CR-030 Detail

violation browser — cell PG port(M1)가 boundary 밖.

■ check_design · CR-030




■ **대상**

mem1/tmp_reg[3] VDD 등 cell PG port 7개, layer M1.

■ **판단**

LVS short / open 0, PG floating 0 → 영향 없음.

# QoR — After route_opt

Issue ① — route_opt 후 tt0p78v125c setup −0.00706 1 path, PT ECO로 진행.

report_qor -summary · after route_opt

**VIOLATED**

```
****************************************
Report : qor
         -summary
         -pba_mode exhaustive
Design : cisc
Version: T-2022.03-SP5
Date   : Wed Oct 28 15:10:55 2026
****************************************
Information: Timer using 'SI, Timing Window Analysis, CRPR'. (TIM-050)

Timing
--------------------------------------------------------------------------------
Context                                WNS            TNS            NVE
--------------------------------------------------------------------------------
normal_ff0p85v125c (Setup)          0.02268        0.00000              0
normal_ff1p16vn40c (Setup)          0.08910        0.00000              0
normal_tt0p78v125c (Setup)         -0.00706       -0.00706              1
normal_tt1p05vn40c (Setup)          0.04319        0.00000              0
Design             (Setup)         -0.00706       -0.00706              1

normal_ff0p85v125c (Hold)           0.08161        0.00000              0
normal_ff1p16vn40c (Hold)           0.03852        0.00000              0
normal_tt0p78v125c (Hold)           0.13795        0.00000              0
normal_tt1p05vn40c (Hold)           0.07940        0.00000              0
Design             (Hold)           0.03852        0.00000              0
--------------------------------------------------------------------------------

Miscellaneous
--------------------------------------------------------------------------------
Cell Area (netlist):                           2253.24070
Cell Area (netlist and physical only):         2253.24100
Nets with DRC Violations:        0
```

**−0.00706**
**Setup WNS (ns)**
tt0p78v125c · 1 path

**+0.03852**
**Hold WNS (ns)**
ff1p16vn40c · NVE 0

**2,253.24**
**Cell area (μm²)**
CTS 2,242.06

**Next**

StarRC SPEF 기반 PrimeTime에서 ECO.

# 06

SECTION

# StarRC & PrimeTime

SMC로 worst · best SPEF 추출, DMSA 4 scenario STA와 fix_eco_timing.

정태웅 · 청주대학교

# star_cmd (1/2)

StarRC 입력 설정 — 추출할 block, ICC2 NDM, layer mapping.

star_cmd_gate · setup

```
***********     SETUP     **********
*** write your block name
* Specify block name for parasitic extraction
BLOCK: cisc_pt

*** write your nlib name
* Provide the input NDM design database
NDM_DATABASE: ../../icc2/nlib/off_route.nlib/

***skip this
* Specify nxtgrd file which consists of capacitance models
TCAD_GRD_FILE:

*** type mapping file which lies in /home/project/DB/32nm_samsung/tech/star_rcxt path
* Provide the mapping file in which design layers mapped to process layers
MAPPING_FILE: /home/project/DB/32nm_samsung/tech/star_rcxt/saed32nm_tf_itf_tluplus.map
```

**BLOCK**
cisc_pt — parasitic 추출 대상 block.

**NDM_DATABASE**
../../icc2/nlib/off_route.nlib/ — ICC2 route nlib.

**MAPPING_FILE**
saed32nm_tf_itf_tluplus.map — design layer ↔ process layer 매핑.

# star_cmd (2/2)

StarRC 출력 설정 — SPEF 형식 · 파일, summary, 작업 directory.

star_cmd_gate · output

```
NETLIST_FORMAT: SPEF

* Provide the name of a file to which output parasitic netlist is written
*** spef file name
NETLIST_FILE: off_cisc.spef


* Provide the name of a summary file to which runtime and memory usage is written
*** STAR-RC summary log file name. in this file you can find error of warning if STAR-RC does not works right.
SUMMARY_FILE: reports/off_cisc_summary.txt

* Provide the working directory name to which StarRC internal information is written in binary
***other detail log file will save at this location.
STAR_DIRECTORY: ./star_dir
```

**NETLIST_FORMAT**
SPEF.

**NETLIST_FILE**
off_cisc.spef — PrimeTime 입력 SPEF.

**SUMMARY_FILE**
reports/off_cisc_summary.txt — error · warning 확인.

**STAR_DIRECTORY**
./star_dir — StarRC 내부 작업 정보.

# smc_config

SMC corner 정의 — worst Cmax 125 °C / best Cmin −40 °C.

smc_config

```
CORNER_NAME: worst
TCAD_GRD_FILE : /home/project/DB/32nm_samsung/tech/star_rcxt/saed32nm_1p9m_Cmax.nxtgrd
OPERATING_TEMPERATURE: 125

CORNER_NAME: best
TCAD_GRD_FILE : /home/project/DB/32nm_samsung/tech/star_rcxt/saed32nm_1p9m_Cmin.nxtgrd
OPERATING_TEMPERATURE: -40
```

**worst**

saed32nm_1p9m_Cmax.nxtgrd · 125 °C → common_*_worst scenario SPEF.

**best**

saed32nm_1p9m_Cmin.nxtgrd · −40 °C → common_*_best scenario SPEF.

# StarRC Summary

message summary — warning 4종, 모두 영향 없음.

StarRC summary

```
Message Summary:
----------------------------------------------------------------------------------------
SX-1813 (x  20)|Warning      |MAPPING_FILE LAYER "()" IS NOT A VALID DATABASE LAYER AND WILL BE REMOVED.
SX-1851 (x   2)|Warning      |LAYER "()" HAS NO CONNECTIONS.
SX-3582 (x   1)|Warning      |COMMAND -CLEAN IS OBSOLETED FOR GATE LEVEL EXTRACTION AND TURNED ON BY DEFAULT.
SX-3860 (x   1)|Warning      |INPUT_NAMES_ESCAPE_REMOVAL IS ENABLED, PLEASE MAKE SURE THAT THE NET SELECT COMMANDS REMOVE THE ESCAPE AT THE SAME TIME.
```

**SX-1813 (20) · SX-1851 (2)**

map file의 빈 "()" layer 항목 — 추출 대상 layer 아님 → 무시.

**SX-3582 (1)**

obsolete 옵션, 기본값 on → 영향 없음.

**SX-3860 (1)**

NETS *로 전체 net 추출 → net 선택 영향 없음.

# PrimeTime DMSA

4 scenario를 DMSA로 동시 분석 — scenario별 common data · SPEF 매핑.

dmsa_run · create_scenario

| Scenario | Common data | SPEF |
|---|---|---|
| normal_ff0p85v125c | common_ff_best | best |
| normal_ff1p16vn40c | common_ff_worst | worst |
| normal_tt0p78v125c | common_tt_worst | worst |
| normal_tt1p05vn40c | common_tt_best | best |




**Session**
pt_shell -multi_scenario, 4 process × 2 core.

**Scenario**
ss 제외 4 scenario, ICC2 출력 netlist · SDC + StarRC SPEF.

**기준**
report_timing · report_constraints -pba_mode exhaustive.

**ECO**
fix_eco_timing 결과를 icctcl로 ICC2에 전달.

# fix_eco_timing

Issue ① — PT 자동 setup ECO, 잔여 위반 endpoint 2개

fix_eco_timing -type setup **VIOLATED**

```
Final ECO Summary:
---------------------------------------------------------
Total number of commands                                0
Total area increased                                 0.00

Fixing Summary:
---------------------------------------------------------
Total violating endpoints found                         2
Total violating endpoints fixed                         0
Total violating endpoints remaining                     2
Total percentage of violations fixed                 0.0%
```

**2**

**Violating endpoints**

found

**0**

**Fixed**

commands 0 · area +0.00

**Next**

자동 ECO로 해결 불가 → 위반 경로 분석.

# Timing Path — After fix_eco_timing

Issue ① — state_out_reg[1] → mem_data2_reg[5] (clk → clk′ half-cycle), slack −0.01142.

report_timing · tt0p78v125c **VIOLATED**

```
Point                                    Fanout    Cap       Incr       Path
-----------------------------------------------------------------------------
clock clk (rise edge)                                       0.00000    0.00000
clock network delay (propagated)                            0.10104    0.10104
top_mc1/state1/state_out_reg[1]/CLK (DFFARX1_LVT)           0.00000    0.10104 r
top_mc1/state1/state_out_reg[1]/QN (DFFARX1_LVT)            0.08753 &  0.18857 r
top_mc1/state1/n7 (net)                        1  1.69014
top_mc1/state1/U3/Y (INVX2_LVT)                             0.01573 &  0.20430 f
top_mc1/state1/state_out[1] (net)              1  3.11748
top_mc1/con_log1/U5/Y (INVX4_LVT)                           0.01699 &  0.22130 r
top_mc1/con_log1/n42 (net)                     3  3.25269
top_mc1/con_log1/U49/Y (OR3X1_LVT)                          0.04687 &  0.26816 r
top_mc1/con_log1/n60 (net)                     1  0.73853
top_mc1/con_log1/U48/Y (AO21X1_LVT)                         0.04378 &  0.31194 r
top_mc1/con_log1/sel (net)                     1  3.05746
top_mc1/addr_mux1/ZINV_185_inst_2132/Y (INVX4_LVT)          0.01781 &  0.32975 f
[illegible]
top_mc1/addr_mux1/ctmTdsLR_1_1061/Y (MUX21X2_LVT)           0.09861 &  0.42835 f
top_mc1/addr_mux1/mux_out[0] (net)            11 15.38619
mem1/U154/Y (OR2X2_LVT)                                     0.07103 &  0.49938 f
[illegible]
mem1/ctmTdsLR_1_775/Y (NOR3X2_LVT)                          0.07066 &  0.57004 r
mem1/n547 (net)                                3  4.14544
mem1/ctmTdsLR_2_1440/Y (AOI22X1_LVT)                        0.06894 &  0.63898 f
mem1/popt_net_550 (net)                        1  1.09078
mem1/ctmTdsLR_1_1439/Y (AND3X1_LVT)                         0.05375 &  0.69273 f
mem1/n423 (net)                                1  3.08556
mem1/ctmTdsLR_2_692/Y (NAND2X0_LVT)                         0.03962 &  0.73235 r
mem1/popt_net_265 (net)                        1  1.00675
mem1/ctmTdsLR_1_2462/Y (OA21X1_LVT)                         0.04597 &  0.77832 r
mem1/n12 (net)                                 1  1.48122
mem1/mem_data2_reg[5]/D (DFFARX1_LVT)                       0.00001 &  0.77833 r
data arrival time                                                      0.77833

clock clk' (rise edge)                                      0.70000    0.70000
clock network delay (propagated)                            0.12713    0.82713
clock reconvergence pessimism                               0.00000    0.82713
clock uncertainty                                          -0.03000    0.79713
mem1/mem_data2_reg[5]/CLK (DFFARX1_LVT)                                0.79713 r
library setup time                                         -0.03022    0.76691
data required time                                                     0.76691
-----------------------------------------------------------------------------
data required time                                                     0.76691
data arrival time                                                     -0.77833
-----------------------------------------------------------------------------
slack (VIOLATED)                                                      -0.01142
```

**−0.01142**
Slack (ns)
HC_P2N

**11 / 15.38619**
mux_out[0]
fanout / cap (fF)

## Path

mux_out[0] → U154 (OR2X2_LVT) → n995 (fanout 5) → mem_data2_reg[5].

## Capture

clk′ rising 0.70 ns — half-cycle path.

# mux_out[0] Net

report_net -verbose — load 11개, U171 (INVX8_LVT) pin cap 최대.

report_net -verbose · tt0p78v125c

```
----------------------------------------
 Verbose output for 'normal_tt0p78v125c'
----------------------------------------

    report_net -connections -verbose [get_nets top_mc1/addr_mux1/mux_out[0]]
****************************************
Report : net
        -connections
        -verbose
Design : cisc
Version: S-2021.06-SP5-2
Date   : Tue Oct 27 14:43:13 2026
****************************************


Connections for net 'top_mc1/addr_mux1/mux_out[0]':
    pin capacitance:    min:7.80795    max:11.0521
    wire capacitance:   min:4.45014*   max:4.45014*  (*:annotated)
    total capacitance:  min:12.2581    max:15.5023
    wire resistance:    min:6.86953e-05*   max:6.86953e-05*  (*:annotated)
    number of drivers:  1
    number of loads:    11
    number of pins:     12

                                                Min       Max
    Driver Pins             Type                Pin Cap   Pin Cap
    ------------            ----------------    --------  --------
    top_mc1/addr_mux1/ctmTdsLR_1_1061/Y Output Pin (MUX21X2_LVT) 0 0

                                                Min       Max
    Load Pins               Type                Pin Cap   Pin Cap
    ------------            ----------------    --------  --------
    mem1/U150/A1            Input Pin (OR3X1_LVT) 0.446543  0.56116
    mem1/U154/A1            Input Pin (OR2X2_LVT) 0.496117  0.625564
    mem1/U68/A2             Input Pin (NAND2X0_RVT) 0.34078 0.473927
    mem1/U171/A             Input Pin (INVX8_LVT) 3.62724   5.45755
    mem1/U4/A2              Input Pin (NAND2X0_RVT) 0.34078 0.473927
    mem1/ctmTdsLR_1_1266/A1 Input Pin (AND2X1_LVT) 0.360315 0.613441
    mem1/U238/A2            Input Pin (OR3X2_LVT) 0.461467  0.606887
    mem1/U142/A2            Input Pin (OR2X2_LVT) 0.46062   0.694987
    mem1/U45/A3             Input Pin (AND3X1_RVT) 0.360756 0.465974
```

**15.5023**
**Total cap max (fF)**
pin 11.0521 + wire 4.45014

**5.45755**
**U171/A max pin cap (fF)**
INVX8_LVT · load 중 최대

**Fix**

U171 → INVX4_LVT로 size down, mux_out[0] 부하 감소.

# U171 Sizing

Issue ① — U171 INVX4_LVT + full effort 후 slack −0.00643 잔여.

report_timing · after U171 sizing **VIOLATED**

```
Startpoint: top_mc1/state1/state_out_reg[1]
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: mem1/mem_data2_reg[5]
            (rising edge-triggered flip-flop clocked by clk')
Last common pin: clk
Path Group: HC_P2N
Path Type: max  (recalculated)
Scenario: normal_tt0p78v125c
Max Data Paths Derating Factor  : 1.05000
Min Clock Paths Derating Factor : 0.92000
Max Clock Paths Derating Factor : 1.05000

Point                                                  Incr       Path
-----------------------------------------------------------------------
clock clk (rise edge)                                0.00000    0.00000
clock network delay (propagated)                     0.10104    0.10104
top_mc1/state1/state_out_reg[1]/CLK (DFFARX1_LVT)    0.00000    0.10104 r
top_mc1/state1/state_out_reg[1]/QN (DFFARX1_LVT)     0.08753 &  0.18857 r
top_mc1/state1/U3/Y (INVX2_LVT)                      0.01573 &  0.20430 f
top_mc1/con_log1/U5/Y (INVX4_LVT)                    0.01699 &  0.22130 r
top_mc1/con_log1/U49/Y (OR3X1_LVT)                   0.04687 &  0.26816 r
top_mc1/con_log1/U48/Y (AO21X1_LVT)                  0.04378 &  0.31194 r
top_mc1/addr_mux1/ZINV_185_inst_2132/Y (INVX4_LVT)   0.01781 &  0.32975 f
top_mc1/addr_mux1/ctmTdsLR_1_1061/Y (MUX21X2_LVT)    0.08828 &  0.41803 r
mem1/U171/Y (INVX4_LVT)                              0.04040 &  0.45843 f
mem1/U57/Y (OR2X2_LVT)                               0.05311 &  0.51154 f
mem1/ctmTdsLR_1_2459/Y (NOR3X2_LVT)                  0.06842 &  0.57996 r
mem1/ctmTdsLR_2_1440/Y (AOI22X1_LVT)                 0.05486 &  0.63482 f
mem1/ctmTdsLR_1_1439/Y (AND3X1_LVT)                  0.05294 &  0.68776 f
mem1/ctmTdsLR_2_692/Y (NAND2X0_LVT)                  0.03959 &  0.72735 r
mem1/ctmTdsLR_1_2462/Y (OA21X1_LVT)                  0.04597 &  0.77332 r
mem1/mem_data2_reg[5]/D (DFFARX1_LVT)                0.00001 &  0.77334 r
data arrival time                                               0.77334

clock clk' (rise edge)                               0.70000    0.70000
clock network delay (propagated)                     0.12713    0.82713
clock reconvergence pessimism                        0.00000    0.82713
clock uncertainty                                   -0.03000    0.79713
mem1/mem_data2_reg[5]/CLK (DFFARX1_LVT)                         0.79713 r
library setup time                                  -0.03022    0.76691
data required time                                              0.76691

data required time                                              0.76691
data arrival time                                              -0.77334
-----------------------------------------------------------------------
slack (VIOLATED)                                               -0.00643
```

## −0.00643

**Slack (ns)**

mem_data2_reg[5]

### Path

mux_out[0] → U171 (INVX4_LVT) → U57 → mem_data2_reg[5].

### Next

PT ECO 한계 → ICC2에서 수동 ECO (clone).

# 07

SECTION

# ECO

PT ECO를 ICC2에 반영하고, clone · Vt swap으로 half-cycle 위반 해소.

# eco_off_route.tcl (1/3)

PT ECO 적용 → MUX clone으로 mux_out[0] fanout 분리.

eco_off_route.tcl · 0–1

```
##############################################################################
# eco_off_route.tcl : post-route ECO after PrimeTime (setup fix) + PT export
#   1) MUX clone      : isolate mem1 half-cycle loads from the mux_out[0] net
#   2) U154 clone     : isolate the ctmTdsLR_1_775/A3 load from the n995 net
#   3) U10 Vt swap    : NOR3 RVT -> LVT
#   4) Save / export  : block cisc_pt_second, netlist, SDC for PT
#   5) Post-ECO checks and reports
##############################################################################

# ----------------------------------------------------------------------------
# 0. Open the routed design
# ----------------------------------------------------------------------------
open_lib nlib/off_route.nlib
open_block cisc.design
link_block

# PT timing ECO (sizing/Vt changes found by fix_eco_timing)
source ../pt/ECO/pt_timing_eco.tcl

# Slack before the clone ECO (reference)
report_qor -summary -pba_mode exhaustive -significant_digits 5
report_timing -scenarios normal_tt0p78v125c -pba_mode exhaustive \
    -significant_digits 5 -nets -capacitance

# ----------------------------------------------------------------------------
# 1. MUX clone : split the mux_out[0] fanout
#    mem1/U154/A1 and mem1/U171/A are moved to the cloned MUX output
# ----------------------------------------------------------------------------
create_cell top_mc1/addr_mux1/mux_clone_a0 [get_lib_cells */MUX21X2_LVT]

connect_net [get_nets top_mc1/addr_mux1/a[0]]       [get_pins top_mc1/addr_mux1/mux_clone_a0/A1]
connect_net [get_nets top_mc1/addr_mux1/b[0]]       [get_pins top_mc1/addr_mux1/mux_clone_a0/A2]
connect_net [get_nets top_mc1/addr_mux1/ZINV_185_2] [get_pins top_mc1/addr_mux1/mux_clone_a0/S0]

# New output port (mux_out_c0) drives the two moved loads
connect_pins -driver [get_pins top_mc1/addr_mux1/mux_clone_a0/Y] \
    -port_name mux_out_c0 -non_incremental \
    [get_pins {mem1/U154/A1 mem1/U171/A}]

# Connectivity check
report_net -connections [get_nets top_mc1/addr_mux1/mux_out_c0]
report_net -connections [get_nets top_mc1/addr_mux1/mux_out[0]]

# Place (new cell, so no -legalize_only), then legalize and route
place_eco_cells -cells [get_cells top_mc1/addr_mux1/mux_clone_a0]
place_eco_cells -eco_changed_cells
check_legality
route_eco
check_routes
```

### PT ECO

pt_timing_eco.tcl source (U171 sizing 포함).

### MUX clone

MUX21X2_LVT mux_clone_a0, 입력 a[0] · b[0] · ZINV_185_2 공유.

### Load 이동

U154/A1 · U171/A → 새 net mux_out_c0.

### 배치

place_eco_cells → route_eco → check_routes.

# eco_off_route.tcl (2/3)

U154 clone → U10 LVT swap.

eco_off_route.tcl · 2–3

```
# ------------------------------------------------------------------------
# 2. U154 clone : drive ctmTdsLR_1_775/A3 from its own OR2 (net n995_c)
# ------------------------------------------------------------------------
create_cell mem1/U154_clone [get_lib_cells */OR2X2_LVT]

connect_net [get_nets mem1/mux_out_c0] [get_pins mem1/U154_clone/A1]
connect_net [get_nets mem1/n476]       [get_pins mem1/U154_clone/A2]

create_net mem1/n995_c
connect_net [get_nets mem1/n995_c] [get_pins mem1/U154_clone/Y]

# Move the load: disconnect by pin only (no -all), then reconnect to the new net
disconnect_net [get_pins mem1/ctmTdsLR_1_775/A3]
connect_net [get_nets mem1/n995_c] [get_pins mem1/ctmTdsLR_1_775/A3]

report_net -connections [get_nets mem1/n995_c]
report_net -connections [get_nets mem1/n995]

place_eco_cells -cells [get_cells mem1/U154_clone]
place_eco_cells -eco_changed_cells
check_legality
route_eco
check_routes

# ------------------------------------------------------------------------
# 3. Vt swap : U10 -> LVT
# ------------------------------------------------------------------------
size_cell [get_cells mem1/U10] [get_lib_cells */NOR3X0_LVT]

# Power/ground for the newly created cells
connect_pg_net

# Timing after ECO
report_qor -summary -pba_mode exhaustive -significant_digits 5
report_timing -scenarios normal_tt0p78v125c -pba_mode exhaustive \
    -significant_digits 5 -slack_lesser_than 0.005 -max_paths 8 -path_type end

# Backup of the ECO result
save_block -as first_eco_off_route
```

**U154 clone**

OR2X2_LVT U154_clone (A1 mux_out_c0 · A2 n476) → 새 net n995_c.

**Load 이동**

ctmTdsLR_1_775/A3만 n995_c로.

**Vt swap**

U10 → NOR3X0_LVT, connect_pg_net.

# eco_off_route.tcl (3/3)

PrimeTime 재확인 + post-ECO check.

eco_off_route.tcl · 4–5

```
# ------------------------------------------------------------------------
# 4. Export for PrimeTime (renamed block, netlist, SDC per scenario)
# ------------------------------------------------------------------------
save_block -as cisc_pt_second
open_block cisc_pt_second.design
current_block cisc_pt_second
link_block

change_names -rules verilog -hierarchy -skip_physical_only_cells \
    -log_changes cisc_names_second.log
write_verilog -exclude {all_physical_cells} outputs_icc2/pt_off_cisc_icc_second.v

foreach sc {normal_ff0p85v125c normal_ff1p16vn40c normal_tt0p78v125c normal_tt1p05vn40c} {
    write_sdc -output outputs_icc2/cisc_${sc}_second.sdc -scenario $sc -nosplit
}

# ------------------------------------------------------------------------
# 5. Post-ECO checks and reports
# ------------------------------------------------------------------------
# Physical checks
check_legality                                          > reports/eco/check_legality_after_off_eco.txt
check_routes                                            > reports/eco/check_routes_after_off_eco.txt
check_lvs -open_reporting detailed -check_child_cells true -max_errors 9999 \
                                                        > reports/eco/check_lvs_after_off_eco.txt
check_pg_drc                                            > reports/eco/check_pg_drc_after_off_eco.txt
check_pg_connectivity                                   > reports/eco/check_pg_connectivity_after_off_eco.txt

# Utilization / Vt ratio
report_utilization                                      > reports/eco/report_utilization_after_off_eco.txt
report_threshold_voltage_groups                         > reports/eco/report_threshold_voltage_groups_after_off_eco.txt

# Timing / constraints
report_qor -summary -significant_digits 5 -pba_mode exhaustive \
                                                        > reports/eco/report_qor_summary_after_off_eco.txt
report_qor -pba_mode exhaustive -significant_digits 5   > reports/eco/report_qor_after_off_eco.txt
report_constraints -all_violators -scenarios [all_scenarios] \
                                                        > reports/eco/report_constraints_after_off_eco.txt
report_timing -delay_type max -scenarios [all_scenarios] -pba_mode exhaustive -significant_digits 5 \
                                                        > reports/eco/report_setup_timing_after_off_eco.txt
report_timing -delay_type min -scenarios [all_scenarios] -pba_mode exhaustive -significant_digits 5 \
                                                        > reports/eco/report_hold_timing_after_off_eco.txt
report_clock_qor                                        > reports/eco/report_clock_qor_after_off_eco.txt
```

**PT 입력**

cisc_pt_second block, 4 scenario SDC.

**Check**

legality · routes · LVS · PG DRC · connectivity, QoR · timing.

# QoR — After PT ECO

Issue ① — ICC2에 PT ECO 반영 후 tt0p78v125c −0.00443 1 path.

report_qor -summary · pba exhaustive

**VIOLATED**

```
Timing
------------------------------------------------------------------------------
Context                                 WNS             TNS             NVE
------------------------------------------------------------------------------
normal_ff0p85v125c (Setup)          0.02270         0.00000               0
normal_ff1p16vn40c (Setup)          0.08910         0.00000               0
normal_tt0p78v125c (Setup)         -0.00443        -0.00443               1
normal_tt1p05vn40c (Setup)          0.04321         0.00000               0
Design             (Setup)         -0.00443        -0.00443               1

normal_ff0p85v125c (Hold)           0.08161         0.00000               0
normal_ff1p16vn40c (Hold)           0.03852         0.00000               0
normal_tt0p78v125c (Hold)           0.13795         0.00000               0
normal_tt1p05vn40c (Hold)           0.07940         0.00000               0
Design             (Hold)           0.03852         0.00000               0
------------------------------------------------------------------------------

Miscellaneous
------------------------------------------------------------------------------
Cell Area (netlist):                           2258.06944
Cell Area (netlist and physical only):         2258.06900
Nets with DRC Violations:        0
```

**−0.00443**

**Setup WNS (ns)**

tt0p78v125c · 1 path

**+0.03852**

**Hold WNS (ns)**

ff1p16vn40c

**2,258.07**

**Cell area (μm²)**

route 2,253.24

# Timing Path — U171 Path

Issue ① — mux_out[0] → U171 (INVX4_LVT) → n689 fanout 19, slack −0.00443.

report_timing · tt0p78v125c **VIOLATED**

```
Startpoint: top_mc1/state1/state_out_reg[1] (rising edge-triggered flip-flop clocked by clk)
Endpoint: mem1/mem_data2_reg[5] (rising edge-triggered flip-flop clocked by clk')
Mode: normal
Corner: tt0p78v125c
Scenario: normal_tt0p78v125c
Path Group: HC_P2N
Path Type: max

Point                                          Fanout      Cap         Incr        Path
-----------------------------------------------------------------------------------------
clock clk (rise edge)                                                  0.00000     0.00000
clock network delay (propagated)                                       0.10101     0.10101

top_mc1/state1/state_out_reg[1]/CLK (DFFARX1_LVT)                      0.00000     0.10101 r
top_mc1/state1/state_out_reg[1]/QN (DFFARX1_LVT)                       0.08713     0.18814 r
top_mc1/state1/n7 (net)                          1      1.62988
top_mc1/state1/U3/Y (INVX2_LVT)                                        0.01564     0.20378 f
top_mc1/state1/state_out[1] (net)                1      3.08266
top_mc1/con_log1/U5/Y (INVX4_LVT)                                      0.01667     0.22045 r
top_mc1/con_log1/n42 (net)                       3      3.08525
top_mc1/con_log1/U49/Y (OR3X1_LVT)                                     0.04639     0.26684 r
top_mc1/con_log1/n60 (net)                       1      0.67131
top_mc1/con_log1/U48/Y (AO21X1_LVT)                                    0.04370     0.31053 r
top_mc1/con_log1/sel (net)                       1      3.05667
top_mc1/addr_mux1/ZINV_185_inst_2132/Y (INVX4_LVT)                     0.01749     0.32802 f
top_mc1/addr_mux1/ZINV_185_2 (net)               5      7.68423
top_mc1/addr_mux1/ctmTdsLR_1_1061/Y (MUX21X2_LVT)                      0.08783     0.41586 r
top_mc1/addr_mux1/mux_out[0] (net)              11     12.63268
mem1/U171/Y (INVX4_LVT)                                                0.03885     0.45471 f
mem1/n689 (net)                                 19     21.98720
mem1/U57/Y (OR2X2_LVT)                                                 0.05270     0.50741 f
mem1/n989 (net)                                  3      3.67768
mem1/ctmTdsLR_1_2459/Y (NOR3X2_LVT)                                    0.06859     0.57600 r
mem1/n926 (net)                                  3      3.61328
mem1/ctmTdsLR_2_1440/Y (AOI22X1_LVT)                                   0.05411     0.63011 f
mem1/popt_net_550 (net)                          1      0.93739
mem1/ctmTdsLR_1_1439/Y (AND3X1_LVT)                                    0.05266     0.68277 f
mem1/n423 (net)                                  1      3.10091
mem1/ctmTdsLR_2_692/Y (NAND2X0_LVT)                                    0.03868     0.72145 r
mem1/popt_net_265 (net)                          1      0.83742
mem1/ctmTdsLR_1_2462/Y (OA21X1_LVT)                                    0.04834     0.76979 r
mem1/n12 (net)                                   1      1.42640
mem1/mem_data2_reg[5]/D (DFFARX1_LVT)                                  0.00003     0.76983 r
data arrival time                                                                  0.76983

clock clk' (rise edge)                                                 0.70000     0.70000
clock network delay (propagated)                                       0.12542     0.82542
clock reconvergence pessimism                                          0.00000     0.82542
mem1/mem_data2_reg[5]/CLK (DFFARX1_LVT)                                0.00000     0.82542 r
clock uncertainty                                                     -0.03000     0.79542
library setup time                                                    -0.03003     0.76540
data required time                                                                 0.76539

data required time                                                                 0.76539
data arrival time                                                                 -0.76983
-----------------------------------------------------------------------------------------
slack (VIOLATED)                                                                  -0.00443
```

**−0.00443**

**Slack (ns)**

mem_data2_reg[5]

**11 / 12.63268**

**mux_out[0]**

fanout / cap (fF)

**19 / 21.98720**

**mem1/n689**

fanout / cap (fF)

# mux_out[0] Loads

report_net -connections — driver ctmTdsLR_1_1061 (MUX21X2_LVT), load 11개.

report_net -connections

```
report_net -connections [get_nets top_mc1/addr_mux1/mux_out[0]]
****************************************
Report : net
         -connections
Design : cisc
Version: T-2022.03-SP5
Date   : Tue Oct 27 13:05:37 2026
****************************************

--------------------------------------------------------------------------------
Reporting for net 'top_mc1/addr_mux1/mux_out[0]'
--------------------------------------------------------------------------------

Connections for net 'top_mc1/addr_mux1/mux_out[0]':

    Driver Pins                   Type
    ------------                  ------------
    top_mc1/addr_mux1/ctmTdsLR_1_1061/Y
                                  Output Pin (MUX21X2_LVT)

    Load Pins                     Type
    ------------                  ------------
    mem1/ctmTdsLR_1_1266/A1       Input Pin (AND2X1_LVT)
    mem1/U45/A3                   Input Pin (AND3X1_RVT)
    mem1/U4/A2                    Input Pin (NAND2X0_RVT)
    mem1/U150/A1                  Input Pin (OR3X1_LVT)
    mem1/U68/A2                   Input Pin (NAND2X0_RVT)
    mem1/U154/A1                  Input Pin (OR2X2_LVT)
    mem1/U142/A2                  Input Pin (OR2X2_LVT)
    mem1/U238/A2                  Input Pin (OR3X2_LVT)
    mem1/U171/A                   Input Pin (INVX4_LVT)
    mem1/U170/A1                  Input Pin (OR3X2_LVT)
    mem1/U178/A3                  Input Pin (NAND3X0_LVT)
```

**Driver**

top_mc1/addr_mux1/ctmTdsLR_1_1061 (MUX21X2_LVT).

**Load**

half-cycle 경로의 U154/A1 · U171/A가 나머지 9 load와 같은 net 공유.

# Load Slack Scan

Issue ① 해결 1 — mux_out[0] load별 slack 확인 → 위반 load만 clone MUX로 분리.

get_timing_paths -through · normal_tt0p78v125c · pba exhaustive

```
icc2_shell> foreach_in_collection p [get_pins -leaf -of_objects [get_nets -of_objects [get_pins top_mc1/addr_mux1/ctmTdsLR_1_1061/Y]] -filter "direction==in"] {
?     puts "[get_object_name $p]  [get_attribute [get_timing_paths -scenarios normal_tt0p78v125c -pba_mode exhaustive -through $p -max_paths 1] slack]"
? }
START_CMD: get_timing_paths    CPU:     34 s ( 0.01 hr) ELAPSE:    584 s ( 0.16 hr) MEM-PEAK:   969 Mb Tue Oct 27 14:01:51 2026
END_CMD: get_timing_paths      CPU:     34 s ( 0.01 hr) ELAPSE:    584 s ( 0.16 hr) MEM-PEAK:   969 Mb Tue Oct 27 14:01:51 2026
mem1/ctmTdsLR_1_1266/A1  0.015322
START_CMD: get_timing_paths    CPU:     34 s ( 0.01 hr) ELAPSE:    584 s ( 0.16 hr) MEM-PEAK:   969 Mb Tue Oct 27 14:01:51 2026
END_CMD: get_timing_paths      CPU:     34 s ( 0.01 hr) ELAPSE:    584 s ( 0.16 hr) MEM-PEAK:   969 Mb Tue Oct 27 14:01:51 2026
mem1/U45/A3  0.018990
START_CMD: get_timing_paths    CPU:     34 s ( 0.01 hr) ELAPSE:    584 s ( 0.16 hr) MEM-PEAK:   969 Mb Tue Oct 27 14:01:51 2026
END_CMD: get_timing_paths      CPU:     34 s ( 0.01 hr) ELAPSE:    584 s ( 0.16 hr) MEM-PEAK:   969 Mb Tue Oct 27 14:01:51 2026
mem1/U4/A2  0.047540
START_CMD: get_timing_paths    CPU:     34 s ( 0.01 hr) ELAPSE:    584 s ( 0.16 hr) MEM-PEAK:   969 Mb Tue Oct 27 14:01:51 2026
END_CMD: get_timing_paths      CPU:     34 s ( 0.01 hr) ELAPSE:    584 s ( 0.16 hr) MEM-PEAK:   969 Mb Tue Oct 27 14:01:52 2026
mem1/U150/A1  0.035089
START_CMD: get_timing_paths    CPU:     34 s ( 0.01 hr) ELAPSE:    584 s ( 0.16 hr) MEM-PEAK:   969 Mb Tue Oct 27 14:01:52 2026
END_CMD: get_timing_paths      CPU:     34 s ( 0.01 hr) ELAPSE:    584 s ( 0.16 hr) MEM-PEAK:   969 Mb Tue Oct 27 14:01:52 2026
mem1/U68/A2  0.030141
START_CMD: get_timing_paths    CPU:     34 s ( 0.01 hr) ELAPSE:    584 s ( 0.16 hr) MEM-PEAK:   969 Mb Tue Oct 27 14:01:52 2026
END_CMD: get_timing_paths      CPU:     35 s ( 0.01 hr) ELAPSE:    584 s ( 0.16 hr) MEM-PEAK:   969 Mb Tue Oct 27 14:01:52 2026
mem1/U154/A1  -0.001984
START_CMD: get_timing_paths    CPU:     35 s ( 0.01 hr) ELAPSE:    584 s ( 0.16 hr) MEM-PEAK:   969 Mb Tue Oct 27 14:01:52 2026
END_CMD: get_timing_paths      CPU:     35 s ( 0.01 hr) ELAPSE:    584 s ( 0.16 hr) MEM-PEAK:   969 Mb Tue Oct 27 14:01:52 2026
mem1/U142/A2  0.022955
START_CMD: get_timing_paths    CPU:     35 s ( 0.01 hr) ELAPSE:    584 s ( 0.16 hr) MEM-PEAK:   969 Mb Tue Oct 27 14:01:52 2026
END_CMD: get_timing_paths      CPU:     35 s ( 0.01 hr) ELAPSE:    584 s ( 0.16 hr) MEM-PEAK:   969 Mb Tue Oct 27 14:01:52 2026
mem1/U238/A2  0.017113
START_CMD: get_timing_paths    CPU:     35 s ( 0.01 hr) ELAPSE:    584 s ( 0.16 hr) MEM-PEAK:   969 Mb Tue Oct 27 14:01:52 2026
END_CMD: get_timing_paths      CPU:     36 s ( 0.01 hr) ELAPSE:    584 s ( 0.16 hr) MEM-PEAK:   969 Mb Tue Oct 27 14:01:52 2026
mem1/U171/A  -0.004434
START_CMD: get_timing_paths    CPU:     36 s ( 0.01 hr) ELAPSE:    584 s ( 0.16 hr) MEM-PEAK:   969 Mb Tue Oct 27 14:01:52 2026
END_CMD: get_timing_paths      CPU:     36 s ( 0.01 hr) ELAPSE:    584 s ( 0.16 hr) MEM-PEAK:   969 Mb Tue Oct 27 14:01:52 2026
mem1/U170/A1  0.017597
START_CMD: get_timing_paths    CPU:     36 s ( 0.01 hr) ELAPSE:    584 s ( 0.16 hr) MEM-PEAK:   969 Mb Tue Oct 27 14:01:52 2026
END_CMD: get_timing_paths      CPU:     37 s ( 0.01 hr) ELAPSE:    584 s ( 0.16 hr) MEM-PEAK:   969 Mb Tue Oct 27 14:01:52 2026
mem1/U178/A3  0.014240
```

**Scan**

mux_out[0] load 11개, load pin별 worst slack 확인.

**Violation**

mem1/U154/A1 −0.001984
mem1/U171/A −0.004434
나머지 9 load slack 양수.

**Fix**

MUX21X2_LVT clone(mux_clone_a0) 생성 → U154/A1 · U171/A만 새 net mux_out_c0로 이동.

# Timing Path — After MUX Clone

Issue ① — clone 후 worst path: n476 → U154 → n995 (fanout 5), slack −0.00677.

report_timing · tt0p78v125c

**VIOLATED**

| Point | Fanout | Cap | Incr | Path |
|---|---|---|---|---|
| clock clk (rise edge) | | | 0.00000 | 0.00000 |
| clock network delay (propagated) | | | 0.10101 | 0.10101 |
| | | | | |
| top_mc1/state1/state_out_reg[1]/CLK (DFFARX1_LVT) | | | 0.00000 | 0.10101 r |
| top_mc1/state1/state_out_reg[1]/QN (DFFARX1_LVT) | | | 0.08711 | 0.18812 r |
| top_mc1/state1/n7 (net) | 1 | 1.62939 | | |
| top_mc1/state1/U3/Y (INVX2_LVT) | | | 0.01562 | 0.20374 f |
| top_mc1/state1/state_out[1] (net) | 1 | 3.07719 | | |
| top_mc1/con_log1/U5/Y (INVX4_LVT) | | | 0.01675 | 0.22049 r |
| top_mc1/con_log1/n42 (net) | 3 | 3.13662 | | |
| top_mc1/con_log1/U49/Y (OR3X1_LVT) | | | 0.04641 | 0.26689 r |
| top_mc1/con_log1/n60 (net) | 1 | 0.67131 | | |
| top_mc1/con_log1/U48/Y (AO21X1_LVT) | | | 0.04370 | 0.31059 r |
| top_mc1/con_log1/sel (net) | 1 | 3.05667 | | |
| top_mc1/addr_mux1/ZINV_185_inst_2132/Y (INVX4_LVT) | | | 0.01963 | 0.33022 f |
| top_mc1/addr_mux1/ZINV_185_2 (net) | 6 | 9.27063 | | |
| top_mc1/addr_mux1/U5/Y (MUX21X2_LVT) | | | 0.07303 | 0.40325 f |
| top_mc1/addr_mux1/mux_out[2] (net) | 1 | 5.47670 | | |
| mem1/ctmTdsLR_1_2361/Y (INVX8_LVT) | | | 0.02729 | 0.43054 r |
| mem1/ZINV_175_6 (net) | 5 | 16.91426 | | |
| [illegible] | | | [illegible] | [illegible] |
| mem1/n476 (net) | 11 | 17.51110 | | |
| mem1/U154/Y (OR2X2_LVT) | | | 0.04999 | 0.49270 f |
| [illegible] | | | [illegible] | [illegible] |
| mem1/ctmTdsLR_1_775/Y (NOR3X2_LVT) | | | 0.06958 | 0.56228 r |
| mem1/n547 (net) | 3 | 3.98391 | | |
| mem1/ctmTdsLR_2_1440/Y (AOI22X1_LVT) | | | 0.06904 | 0.63132 f |
| mem1/popt_net_550 (net) | 1 | 0.94122 | | |
| mem1/ctmTdsLR_1_1439/Y (AND3X1_LVT) | | | 0.05417 | 0.68549 f |
| mem1/n423 (net) | 1 | 3.23542 | | |
| mem1/ctmTdsLR_2_692/Y (NAND2X0_LVT) | | | 0.03837 | 0.72387 r |
| mem1/popt_net_265 (net) | 1 | 0.83742 | | |
| mem1/ctmTdsLR_1_2462/Y (OA21X1_LVT) | | | 0.04828 | 0.77215 r |
| mem1/n12 (net) | 1 | 1.42640 | | |
| mem1/mem_data2_reg[5]/D (DFFARX1_LVT) | | | 0.00003 | 0.77219 r |
| data arrival time | | | | 0.77219 |
| | | | | |
| clock clk' (rise edge) | | | 0.70000 | 0.70000 |
| clock network delay (propagated) | | | 0.12546 | 0.82546 |
| clock reconvergence pessimism | | | 0.00000 | 0.82546 |
| mem1/mem_data2_reg[5]/CLK (DFFARX1_LVT) | | | 0.00000 | 0.82546 r |
| clock uncertainty | | | -0.03000 | 0.79546 |
| library setup time | | | -0.03004 | 0.76542 |
| data required time | | | | 0.76542 |
| data required time | | | | 0.76542 |
| data arrival time | | | | -0.77219 |
| slack (VIOLATED) | | | | -0.00677 |

**−0.00677**

**Slack (ns)**

mem_data2_reg[5]

**11 / 17.51110**

**mem1/n476**

fanout / cap (fF)

**5 / 6.90768**

**mem1/n995**

fanout / cap (fF)

# U154 Through Paths

report_timing -through U154/Y — 위반은 mem_data2_reg[5] 1개.

report_timing -through mem1/U154/Y · end

```
icc2_shell> report_timing -scenarios normal_tt0p78v125c -pba_mode exhaustive -significant_digits 5 \
?     -through [get_pins mem1/U154/Y] -max_paths 5 -path_type end
START_CMD: report_timing       CPU:     54 s ( 0.01 hr) ELAPSE:    991 s ( 0.28 hr) MEM-PEAK:  1282 Mb Tue Oct 27 13:40:04 2026
****************************************
Report : timing
        -path_type end
        -delay_type max
        -max_paths 5
        -report_by design
        -pba_mode exhaustive
Design : cisc
Version: T-2022.03-SP5
Date   : Tue Oct 27 13:40:04 2026
****************************************
Information: Timer using 'SI, Timing Window Analysis, CRPR'. (TIM-050)
Endpoint                          Path Delay     Path Required       CRP    Slack Scenario
-------------------------------------------------------------------------------------------
mem1/mem_data2_reg[5]/D (DFFARX1_LVT)
                                  0.77219 r      0.76542        0.00000
                                                                        -0.00677
                                                                               normal_tt0p78v125c
mem1/mem_data2_reg[4]/D (DFFARX1_LVT)
                                  0.75691 r      0.76402        0.00000
                                                                         0.00711
                                                                               normal_tt0p78v125c
mem1/mem_data2_reg[6]/D (DFFARX1_LVT)
                                  0.75655 r      0.76426        0.00000
                                                                         0.00771
                                                                               normal_tt0p78v125c
mem1/tmp_reg[0]/D (DFFX1_RVT)     0.79575 r      0.80655        0.00000
                                                                         0.01080
                                                                               normal_tt0p78v125c
mem1/tmp_reg[4]/D (DFFX1_RVT)     0.79527 r      0.80623        0.00000
                                                                         0.01095
                                                                               normal_tt0p78v125c
END_CMD: report_timing         CPU:     54 s ( 0.01 hr) ELAPSE:    991 s ( 0.28 hr) MEM-PEAK:  1282 Mb Tue Oct 27 13:40:04 2026
```

**Violation**

mem_data2_reg[5] −0.00677.

**Met**

reg[4] +0.00711 · reg[6] +0.00771
tmp_reg[0] +0.01080 · tmp_reg[4] +0.01095

**Fix**

ctmTdsLR_1_775/A3만 OR2X2_LVT clone(U154_clone)으로 분리.

# Timing Path — After U154 Clone

Issue ① — 위반 해소, worst path mem_data2_reg[4] +0.00092 — U10 LVT로 마진 확보.

report_timing · tt0p78v125c

MET

```
Startpoint: top_mc1/state1/state_out_reg[1] (rising edge-triggered flip-flop clocked by clk)
Endpoint: mem1/mem_data2_reg[4] (rising edge-triggered flip-flop clocked by clk')
Mode: normal
Corner: tt0p78v125c
Scenario: normal_tt0p78v125c
Path Group: HC_P2N
Path Type: max

Point                                          Incr         Path
-----------------------------------------------------------------------------
clock clk (rise edge)                        0.00000      0.00000
clock network delay (propagated)             0.10101      0.10101

top_mc1/state1/state_out_reg[1]/CLK (DFFARX1_LVT)
                                             0.00000      0.10101 r
top_mc1/state1/state_out_reg[1]/QN (DFFARX1_LVT)
                                             0.08711      0.18812 r
top_mc1/state1/U3/Y (INVX2_LVT)              0.01562      0.20374 f
top_mc1/con_log1/U5/Y (INVX4_LVT)            0.01661      0.22035 r
top_mc1/con_log1/U49/Y (OR3X1_LVT)           0.04639      0.26674 r
top_mc1/con_log1/U48/Y (AO21X1_LVT)          0.04370      0.31044 r
top_mc1/addr_mux1/ZINV_185_inst_2132/Y (INVX4_LVT)
                                             0.01951      0.32995 f
top_mc1/addr_mux1/U4/Y (MUX21X2_LVT)         0.06952      0.39947 r
mem1/ctmTdsLR_1_2336/Y (INVX8_LVT)           0.03128      0.43075 f
mem1/U222/Y (INVX16_LVT)                     0.01860      0.44935 r
mem1/U238/Y (OR3X2_LVT)                      0.06086      0.51021 r
mem1/U10/Y (NOR3X0_RVT)                      0.09410      0.60431 f
mem1/U372/Y (NOR3X0_LVT)                     0.05914      0.66345 f
mem1/U90/Y (NAND4X0_LVT)                     0.03361      0.69706 f
mem1/ctmTdsLR_1_1093/Y (MUX21X2_LVT)         0.06603      0.76309 f
mem1/mem_data2_reg[4]/D (DFFARX1_LVT)        0.00003      0.76312 f
data arrival time                                         0.76312

clock clk' (rise edge)                       0.70000      0.70000
clock network delay (propagated)             0.12552      0.82552
clock reconvergence pessimism                0.00000      0.82552
mem1/mem_data2_reg[4]/CLK (DFFARX1_LVT)      0.00000      0.82552 r
clock uncertainty                           -0.03000      0.79552
library setup time                          -0.03147      0.76405
data required time                                        0.76404
-----------------------------------------------------------------------------
data required time                                        0.76404
data arrival time                                        -0.76312
-----------------------------------------------------------------------------
slack (MET)                                               0.00092
```

**+0.00092**

**Slack (ns)**

mem_data2_reg[4]

### U10

경로 위 NOR3X0_RVT (incr 0.09410) → NOR3X0_LVT swap.

### 목적

half-cycle 경로 setup 마진 확보.

# QoR — After ECO

Issue ① 해결 — 4 scenario setup · hold 위반 0.

report_qor -summary · after U10 swap

RESOLVED

```
icc2_shell> report_qor -summary -significant_digits 5 -pba_mode exhaustive
****************************************
Report : qor
         -summary
         -pba_mode exhaustive
Design : cisc
Version: T-2022.03-SP5
Date   : Tue Oct 27 13:48:31 2026
****************************************
Information: Timer using 'SI, Timing Window Analysis, CRPR'. (TIM-050)

Timing
--------------------------------------------------------------------------------
Context                                WNS            TNS            NVE
--------------------------------------------------------------------------------
normal_ff0p85v125c (Setup)          0.02262        0.00000              0
normal_ff1p16vn40c (Setup)          0.08908        0.00000              0
normal_tt0p78v125c (Setup)          0.00032        0.00000              0
normal_tt1p05vn40c (Setup)          0.04314        0.00000              0
Design             (Setup)          0.00032        0.00000              0

normal_ff0p85v125c (Hold)           0.08163        0.00000              0
normal_ff1p16vn40c (Hold)           0.03854        0.00000              0
normal_tt0p78v125c (Hold)           0.13794        0.00000              0
normal_tt1p05vn40c (Hold)           0.07940        0.00000              0
Design             (Hold)           0.03854        0.00000              0
--------------------------------------------------------------------------------

Miscellaneous
--------------------------------------------------------------------------------
Cell Area (netlist):                          2263.91475
Cell Area (netlist and physical only):        2263.91500
Nets with DRC Violations:        0
```

**+0.00032**

**Setup WNS (ns)**

tt0p78v125c · NVE 0

**+0.03854**

**Hold WNS (ns)**

ff1p16vn40c

**2,263.91**

**Cell area (μm²)**

ECO 전 2,258.07

**0**

**Nets with DRC**

# 2nd PT — Setup

1차 ECO 결과로 StarRC · DMSA 재분석 → fix_eco_timing 후 setup 위반 path 없음.

report_timing -delay_type max **PASS**

```
pt_shell> report_timing \
?     -delay_type max \
?     -max_paths 20 \
?     -significant_digits 5 \
?     -pba_mode exhaustive
****************************************
Report : timing
        -path_type full
        -delay_type max
        -slack_lesser_than 0.00000
        -max_paths 20
        -pba_mode exhaustive
        -sort_by slack
Design : multi-scenario design
Version: S-2021.06-SP5-2
Date   : Tue Oct 27 15:37:01 2026
****************************************


Start of Master/Worker Task Processing
-----------------------------------
Started    : Command execution in scenario 'normal_tt0p78v125c'
Started    : Command execution in scenario 'normal_tt1p05vn40c'
Started    : Command execution in scenario 'normal_ff0p85v125c'
Started    : Command execution in scenario 'normal_ff1p16vn40c'
Succeeded  : Command execution in scenario 'normal_ff0p85v125c'
Succeeded  : Command execution in scenario 'normal_tt1p05vn40c'
Succeeded  : Command execution in scenario 'normal_ff1p16vn40c'
Succeeded  : Command execution in scenario 'normal_tt0p78v125c'
-----------------------------------
End of Master/Worker Task Processing

No paths with slack less than 0.00000.
```

**No paths**

**Setup slack < 0**

4 scenario

**4 / 4**

**Scenario succeeded**

ff0p85 · ff1p16 · tt0p78 · tt1p05

# 2nd PT — Hold

DMSA 4 scenario — hold 위반 path 없음.

report_timing -delay_type min **PASS**

```
pt_shell> report_timing \
?      -delay_type min \
?      -max_paths 20 \
?      -significant_digits 5 \
?      -pba_mode exhaustive
****************************************
Report : timing
         -path_type full
         -delay_type min
         -slack_lesser_than 0.00000
         -max_paths 20
         -pba_mode exhaustive
         -sort_by slack
Design : multi-scenario design
Version: S-2021.06-SP5-2
Date   : Tue Oct 27 15:37:01 2026
****************************************


Start of Master/Worker Task Processing
----------------------------------
Started    : Command execution in scenario 'normal_tt0p78v125c'
Started    : Command execution in scenario 'normal_ff0p85v125c'
Started    : Command execution in scenario 'normal_tt1p05vn40c'
Started    : Command execution in scenario 'normal_ff1p16vn40c'
Succeeded  : Command execution in scenario 'normal_ff0p85v125c'
Succeeded  : Command execution in scenario 'normal_tt0p78v125c'
Succeeded  : Command execution in scenario 'normal_ff1p16vn40c'
Succeeded  : Command execution in scenario 'normal_tt1p05vn40c'
----------------------------------
End of Master/Worker Task Processing

No paths with slack less than 0.00000.
```

## No paths

**Hold slack < 0**

4 scenario

## 4 / 4

**Scenario succeeded**

ff0p85 · ff1p16 · tt0p78 · tt1p05

# 2nd PT — Constraints

report_constraints -all_violators — 위반 항목 없음 → 2차 ECO tcl을 ICC2에 반영.

report_constraints -all_violators · pba exhaustive **PASS**

```
pt_shell> report_constraints -all_violators -significant_digits 5 -pba_mode exhaustive

Start of Master/Worker Task Processing
-------------------------------------
Started    : Command execution in scenario 'normal_ff0p85v125c'
Started    : Command execution in scenario 'normal_tt0p78v125c'
Started    : Command execution in scenario 'normal_ff1p16vn40c'
Started    : Command execution in scenario 'normal_tt1p05vn40c'
Succeeded  : Command execution in scenario 'normal_ff0p85v125c'
Succeeded  : Command execution in scenario 'normal_tt0p78v125c'
Succeeded  : Command execution in scenario 'normal_ff1p16vn40c'
Succeeded  : Command execution in scenario 'normal_tt1p05vn40c'
-------------------------------------
End of Master/Worker Task Processing

****************************************
Report : constraint
        -all_violators
        -path slack_only
        -ignore_register_feedback 0.00
        -pba_mode exhaustive
Design : multi_scenario design
Version: S-2021.06-SP5-2
Date   : Tue Oct 27 15:37:01 2026
****************************************

Scenarios: normal_ff0p85v125c normal_ff1p16vn40c normal_tt0p78v125c normal_tt1p05vn40c
```

**0**

**Violators**

4 scenario

**Next**

pt_timing_eco_second.tcl을 1차 ECO block에 적용.

# eco_off_route_second.tcl (1/2)

2차 PT ECO 적용 → route error 수동 수정 → physical check → 최종 PT 입력.

eco_off_route_second.tcl · 1–3

```
##########################################################################
# off_eco_route_second.tcl : final fixing in ICC2 after the second PT ECO, then fillers
#   1) Apply the PT ECO (second) and fix the remaining route errors
#   2) Physical checks
#   3) Export netlist / SDC for the PT re-check (block cisc_pt_final)
#   4) Filler insertion (after all ECO is finished)
#   5) Final checks and reports, save nlib/off_after_filler.nlib
##########################################################################

# ------------------------------------------------------------------------
# 1. Open the first ECO result and apply the second PT ECO
# ------------------------------------------------------------------------
open_lib nlib/off_route.nlib
open_block first_eco_off_route.design
link_block

source ../pt/ECO/pt_timing_eco_second.tcl

# Route errors left by the ECO are fixed manually
# after manual fixing
check_routes
route_eco -open_net_driven true

# ------------------------------------------------------------------------
# 2. Physical checks after ECO
# ------------------------------------------------------------------------
check_legality
check_routes
check_lvs -open_reporting detailed -check_child_cells true -max_errors 9999
check_pg_drc
check_pg_connectivity
check_pg_missing_vias

# ------------------------------------------------------------------------
# 3. Export for the PT re-check (ICC2 fixed the PT violations, so PT must
#    confirm again with a new netlist, SDC and SPEF)
#    The block is saved first so the ECO result is kept, then a renamed
#    copy is exported.
# ------------------------------------------------------------------------
save_block -as second_eco_off_route
save_block -as cisc_pt_final
open_block cisc_pt_final.design
current_block cisc_pt_final
link_block
change_names -rules verilog -hierarchy -skip_physical_only_cells \
    -log_changes cisc_names_final.log
write_verilog -exclude {all_physical_cells} outputs_icc2/pt_off_cisc_icc_final.v

foreach sc {normal_ff0p85v125c normal_ff1p16vn40c normal_tt0p78v125c normal_tt1p05vn40c} {
    write_sdc -output outputs_icc2/cisc_${sc}_final.sdc -scenario $sc -nosplit
}
```

**PT ECO**

1차 ECO block에 pt_timing_eco_second.tcl source.

**Route**

route error 수동 수정 → route_eco -open_net_driven true.

**Check**

legality · routes · LVS · PG DRC · connectivity · missing via.

**PT 입력**

cisc_pt_final block, 4 scenario SDC.

# eco_off_route_second.tcl (2/2)

모든 ECO 완료 후 filler 삽입 → 최종 check.

eco_off_route_second.tcl · 4–5

```
# ------------------------------------------------------------------------
# 4. Filler insertion on the original (non-renamed) block
#    SHFILL*_RVT only: excludes DHFILL (double height), LVT and HVT fillers
# ------------------------------------------------------------------------
open_block second_eco_off_route.design
current_block second_eco_off_route
link_block

set FILLERS [get_lib_cells */SHFILL*_RVT]
get_object_name $FILLERS

create_stdcell_fillers -lib_cells $FILLERS
connect_pg_net
remove_stdcell_fillers_with_violation

# If check_legality fails, remove the fillers and insert again:
#   remove_cells [get_cells -hierarchical -filter "ref_name=~*FILL*"]

# ------------------------------------------------------------------------
# 5. Final checks and reports
# ------------------------------------------------------------------------
check_legality                                      > reports/eco/check_legality_after_off_filler.txt
check_routes                                        > reports/eco/check_routes_after_off_filler.txt
check_lvs -open_reporting detailed -check_child_cells true -max_errors 9999 \
                                                    > reports/eco/check_lvs_after_off_filler.txt
check_pg_drc                                        > reports/eco/check_pg_drc_after_off_filler.txt
check_pg_connectivity                               > reports/eco/check_pg_connectivity_after_off_filler.txt
check_pg_missing_vias                               > reports/eco/check_pg_missing_vias_after_off_filler.txt

report_utilization                                  > reports/eco/report_utilization_after_off_filler.txt
report_threshold_voltage_groups                     > reports/eco/report_threshold_voltage_groups_after_off_filler.txt
report_qor -summary -significant_digits 5 -pba_mode exhaustive \
                                                    > reports/eco/report_qor_summary_after_off_filler.txt
report_qor -pba_mode exhaustive -significant_digits 5   > reports/eco/report_qor_after_off_filler.txt
report_constraints -all_violators -scenarios [all_scenarios] \
                                                    > reports/eco/report_constraints_after_off_filler.txt
report_timing -delay_type max -scenarios [all_scenarios] -pba_mode exhaustive -significant_digits 5 \
                                                    > reports/eco/report_setup_timing_after_off_filler.txt
report_timing -delay_type min -scenarios [all_scenarios] -pba_mode exhaustive -significant_digits 5 \
                                                    > reports/eco/report_hold_timing_after_off_filler.txt
report_clock_qor                                    > reports/eco/report_clock_qor_after_off_filler.txt

save_lib -as nlib/off_after_filler.nlib
save_block -as final_route
```

**Filler**

SHFILL*_RVT만 사용 (DHFILL · LVT · HVT filler 제외).

**PG**

connect_pg_net → 위반 filler 제거 (remove_stdcell_fillers_with_violation).

**Check**

legality · routes · LVS · PG, QoR · timing · clock QoR.

# check_routes — After 2nd ECO

Issue ② — 2차 PT ECO 반영 후 route DRC 10.

check_routes · after 2nd PT ECO

**VIOLATED**

```
Verify Summary:

Total number of nets = 932, of which 0 are not extracted
Total number of open nets = 0, of which 0 are frozen
Total number of excluded ports = 0 ports of 0 unplaced cells connected to 0 nets
                                 0 ports without pins of 0 cells connected to 0 nets
                                 0 ports of 0 cover cells connected to 0 non-pg nets
Total number of DRCs = 10
Total number of antenna violations = no antenna rules defined
Total number of tie to rail violations = not checked
Total number of tie to rail directly violations = not checked


END_CMD: check_routes          CPU:     19 s ( 0.01 hr) ELAPSE:     17 s ( 0.00 hr) M
```

**10**
DRCs
zroute.err

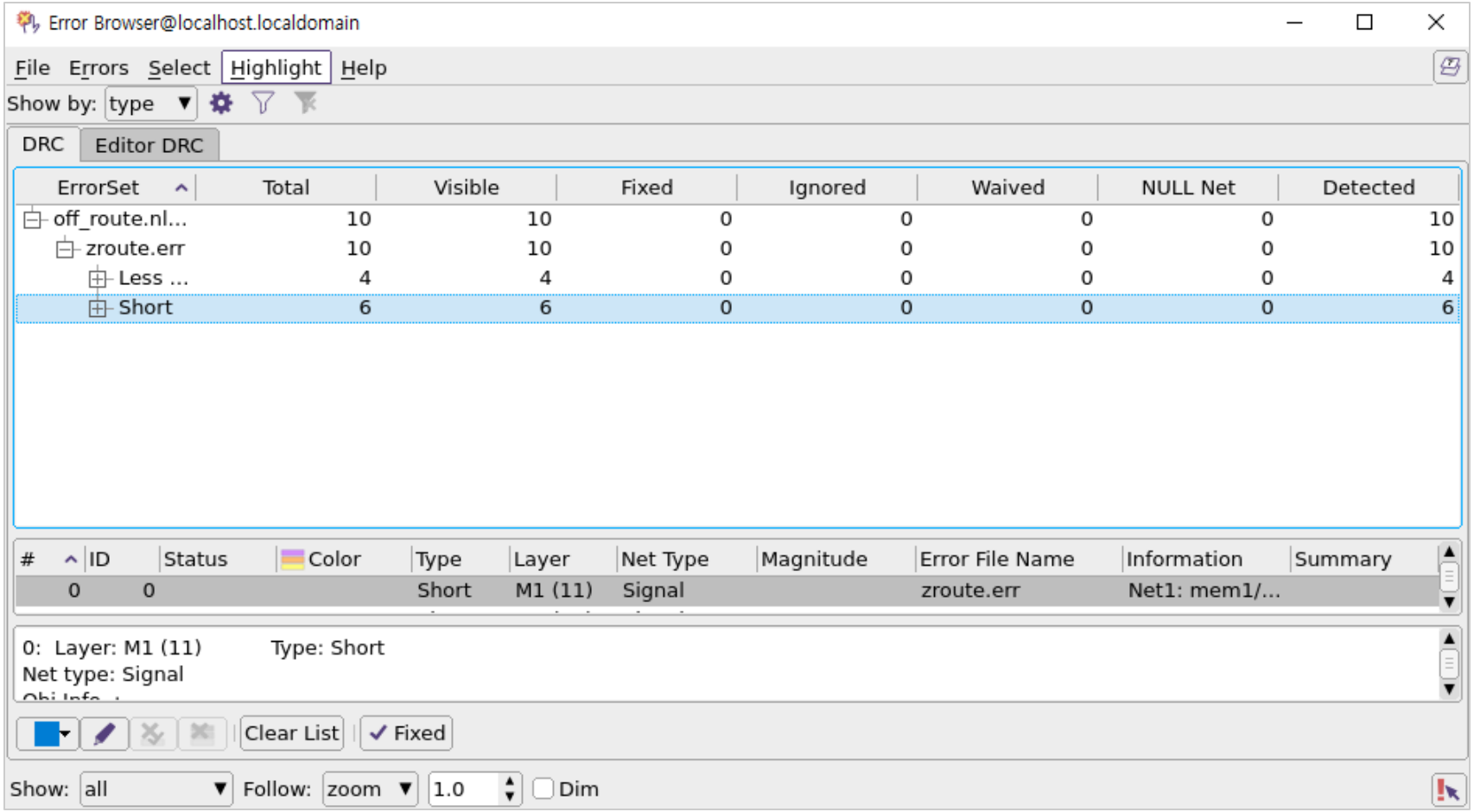
**0**
Open nets

**932**
Nets

# Error Browser

Issue ② — zroute.err: Short 6 · Less than minimum enclosed area 4, M1.

Error Browser · zroute.err




**6**

**Short**

M1 · Signal / Power / Ground

**4**

**Less than minimum enclosed area**

M1 · Signal

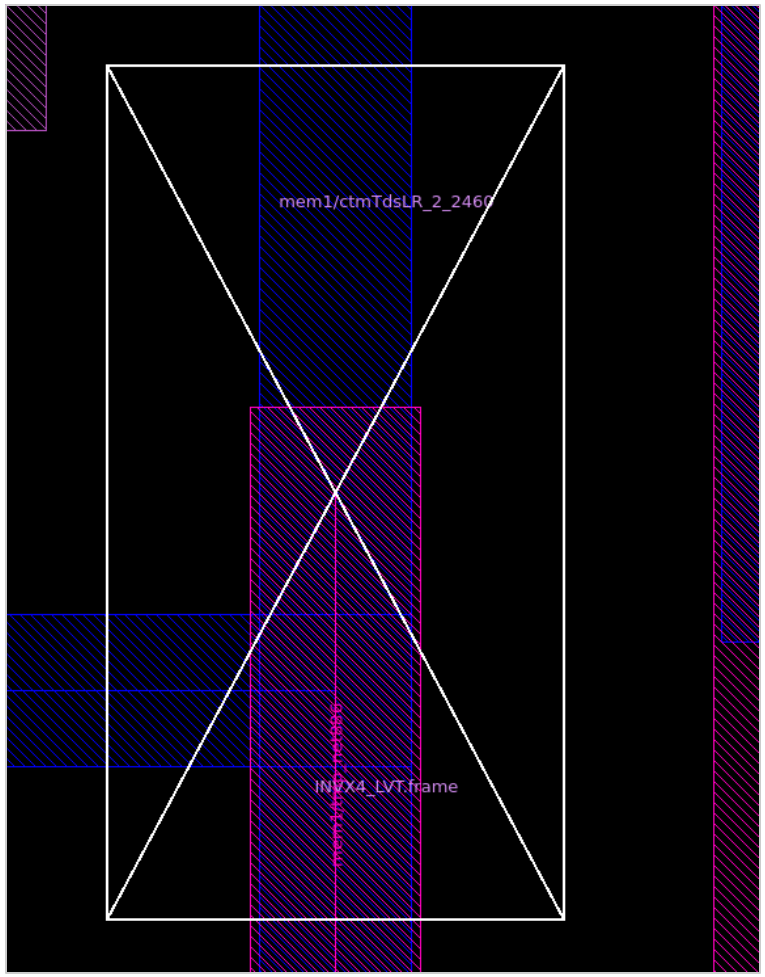
**Cause**

2차 PT ECO로 바뀐 cell 주변 M1 배선.

# Manual Fix

Issue ② — ECO cell 주변 M1 route error를 수동으로 지워 재배선.

layout · mem1/ctmTdsLR_2_2460 · INVX4_LVT




**Fix**

error net, via를 수동으로 삭제 → route_eco -open_net_driven true로 재배선.

**대상**

2차 PT ECO로 바뀐 cell 주변 M1 signal / Power / Ground (Short · Less …).

# check_routes — Re-check

Issue ② 해결 — route DRC 10 → 0.

check_routes · after manual fix

RESOLVED

```
Verify Summary:

Total number of nets = 932, of which 0 are not extracted
Total number of open nets = 0, of which 0 are frozen
Total number of excluded ports = 0 ports of 0 unplaced cells connected to 0 nets
                                 0 ports without pins of 0 cells connected to 0 nets
                                 0 ports of 0 cover cells connected to 0 non-pg nets
Total number of DRCs = 0
Total number of antenna violations = no antenna rules defined
Total number of tie to rail violations = not checked
Total number of tie to rail directly violations = not checked


END_CMD: check_routes          CPU:     88 s ( 0.02 hr) ELAPSE:    506 s ( 0.14 hr) M
```

**10 → 0**
DRCs
수동 수정

**0**
Open nets

**932**
Nets

# 08

SECTION

# Filler & Sign-off

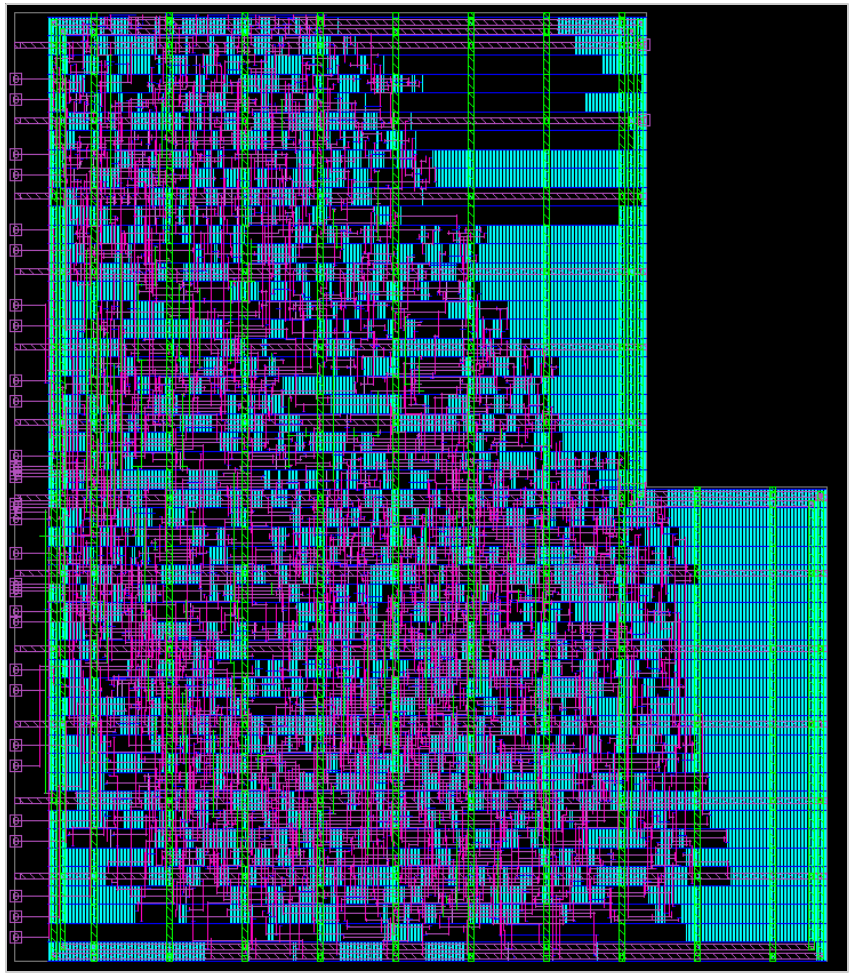
모든 ECO 이후 filler 삽입, DRC · LVS · PG · timing과 최종 PT 확인.

정태웅 · 청주대학교

# Filler Insertion

빈 site를 SHFILL*_RVT로 채운 최종 layout.

layout · after filler



**9,511**

**Filler cells**

physical-only · SHFILL*_RVT

**2,265.44**

**Cell area (μm²)**

netlist

**5,069.92**

**Cell area (μm²)**

netlist + physical-only

# check_routes — Final

filler 삽입 후 DRC 0 · open net 0.

check_routes · after filler

PASS

```
Verify Summary:

Total number of nets = 932, of which 0 are not extracted
Total number of open nets = 0, of which 0 are frozen
Total number of excluded ports = 0 ports of 0 unplaced cells connected to 0 nets
                                 0 ports without pins of 0 cells connected to 0 nets
                                 0 ports of 0 cover cells connected to 0 non-pg nets
Total number of DRCs = 0
Total number of antenna violations = no antenna rules defined
Total number of tie to rail violations = not checked
Total number of tie to rail directly violations = not checked


END_CMD: check_routes          CPU:     35 s ( 0.01 hr) ELAPSE:    193 s ( 0.05 hr)
```

**0**
DRCs

**0**
Open nets

**932**
Nets
0 not extracted

# Legality · LVS — Final

check_legality succeeded · check_lvs short · open · floating 0.

check_legality **SUCCEEDED**

```
check_legality for block design cisc succeeded!


check_legality succeeded.

**************************

END_CMD: check_legality        CPU:     31 s ( 0
```

check_lvs **PASS**

```
Total number of input nets is 932.
Total number of short violations is 0.
Total number of open locations is 0 (0 open nets).
Total number of floating route violations is 0.

Elapsed =    0:00:01, CPU =    0:00:02
END_CMD: check_lvs             CPU:     40 s ( 0.0
```

**SUCCEEDED**

**Legality**
filler 포함

**0 / 0 / 0**

**LVS**
short / open / floating

**932**

**Input nets**

# PG Check — Final

check_pg_connectivity · check_pg_drc · check_pg_missing_vias — 위반 없음.

### check_pg_connectivity

```
**************Verify net VDD connectivity*****************
  Number of floating wires: 0
  Number of floating vias: 0
  Number of floating std cells: 0
  Number of floating hard macros: 0
  Number of floating I/O pads: 0
  Number of floating terminals: 0
  Number of floating hierarchical blocks: 0
************************************************************
Loading cell instances...
Loading P/G wires and vias...
Number of VSS Wires: 43
Number of VSS Vias: 394
Number of VSS Terminals: 22
**************Verify net VSS connectivity*****************
  Number of floating wires: 0
  Number of floating vias: 0
  Number of floating std cells: 0
  Number of floating hard macros: 0
  Number of floating I/O pads: 0
  Number of floating terminals: 0
  Number of floating hierarchical blocks: 0
************************************************************
Overall runtime: 0 seconds.
END_CMD: check_pg_connectivity CPU:     57 s ( 0.02 hr) ELAPS
```

### check_pg_drc · missing_vias

PASS

```
No errors found.
END_CMD: check_pg_drc                CPU:
icc2_shell> check_pg_missing_vias
Information: The command 'check_pg_mi
START_CMD: check_pg_missing_vias CPU:
Check net VDD vias...
Number of missing vias: 0
Checking net VDD vias took 0 seconds.
Check net VSS vias...
Number of missing vias: 0
Checking net VSS vias took 0 seconds.
Overall runtime: 0 seconds.
END_CMD: check_pg_missing_vias CPU:
```

**0**

**Floating**

VDD · VSS 전 항목

**43 / 394**

**VSS wires / vias**

terminal 22

**0 / 0**

**Missing vias**

VDD / VSS · PG DRC no errors

# QoR Summary — Final

PBA exhaustive 기준 4 scenario setup · hold 위반 0.

report_qor -summary · after filler

0 NVE

```
****************************************
Report : qor
         -summary
         -pba_mode exhaustive
Design : cisc
Version: T-2022.03-SP5
Date   : Wed Oct 28 16:57:40 2026
****************************************
Information: Timer using 'SI, Timing Window Analysis, CRPR'. (TIM-050)

Timing
--------------------------------------------------------------------------------
Context                                WNS             TNS             NVE
--------------------------------------------------------------------------------
normal_ff0p85v125c (Setup)          0.02260         0.00000               0
normal_ff1p16vn40c (Setup)          0.08907         0.00000               0
normal_tt0p78v125c (Setup)          0.00027         0.00000               0
normal_tt1p05vn40c (Setup)          0.04312         0.00000               0
Design             (Setup)          0.00027         0.00000               0

normal_ff0p85v125c (Hold)           0.08163         0.00000               0
normal_ff1p16vn40c (Hold)           0.03854         0.00000               0
normal_tt0p78v125c (Hold)           0.13797         0.00000               0
normal_tt1p05vn40c (Hold)           0.07940         0.00000               0
Design             (Hold)           0.03854         0.00000               0
--------------------------------------------------------------------------------

Miscellaneous
--------------------------------------------------------------------------------
Cell Area (netlist):                           2265.43962
Cell Area (netlist and physical only):         5069.91900
Nets with DRC Violations:       0
```

**+0.00027**
Setup WNS (ns)
tt0p78v125c

**+0.03854**
Hold WNS (ns)
ff1p16vn40c

**2,265.44**
Cell area (μm²)
netlist · +physical 5,069.92

**0**
Nets with DRC

# Utilization — Final

report_utilization — 0.4451.

report_utilization · after filler

```
****************************************
Report : report_utilization
Design : cisc
Version: T-2022.03-SP5
Date   : Wed Oct 28 16:55:34 2026
****************************************
Utilization Ratio:                      0.4451
Utilization options:
 - Area calculation based on:           site_row of block final_route
 - Categories of objects excluded:      hard_macros macro_keepouts soft_macros io_cells hard_blockages
Total Area:                             5089.2336
Total Capacity Area:                    5089.2336
Total Area of cells:                    2265.4396
Area of excluded objects:
 - hard_macros          :               0.0000
 - macro_keepouts       :               0.0000
 - soft_macros          :               0.0000
 - io_cells             :               0.0000
 - hard_blockages       :               0.0000

Utilization of site-rows with:
 - Site 'unit':                         0.4451

0.4451
```

**0.4451**
**Utilization**
site row 기준

**2,265.44**
**Cell area (μm²)**
netlist

**추이**
floorplan 0.4383 → place 0.4584 → CTS 0.4406 → route 0.4427 → ECO 0.4448 → final 0.4451.

# Vth Ratio — Final

표기상 LVT 2.40 %는 filler 포함 — logic cell 기준 28.30 %.

report_threshold_voltage_groups · after filler

```
****************************************
Report : threshold_voltage_group
Design : cisc
Version: T-2022.03-SP5
Date   : Wed Oct 28 16:56:07 2026
****************************************

Cell Count Report
-----------------------------------------------------------------------------------------------------------------------------------------------------------------------

Group Names                    All (%)       Repeaters (%)   Combinational (%)       Registers (%)     Sequentials (%)   Clock_network (%)          Macro (%)   Physical_only (%)
-----------------------------------------------------------------------------------------------------------------------------------------------------------------------
saed32cell_hvt             232   (2.23%)       104   (1.00%)       127   (1.22%)         1   (0.01%)         0   (0.00%)         0   (0.00%)         0   (0.00%)         0   (0.00%)
saed32cell_lvt             249   (2.40%)        55   (0.53%)       179   (1.72%)        15   (0.14%)         0   (0.00%)         0   (0.00%)         0   (0.00%)         0   (0.00%)  L
saed32cell_svt            9910  (95.37%)        80   (0.77%)       243   (2.34%)        74   (0.71%)         0   (0.00%)         2   (0.02%)         0   (0.00%)      9511  (91.53%)
-----------------------------------------------------------------------------------------------------------------------------------------------------------------------
Total                    10391 (100.00%)       239   (2.30%)       549   (5.28%)        90   (0.87%)         0   (0.00%)         2   (0.02%)         0   (0.00%)      9511  (91.53%)
Low vth groups             249   (2.40%)        55   (0.53%)       179   (1.72%)        15   (0.14%)         0   (0.00%)         0   (0.00%)         0   (0.00%)         0   (0.00%)
-----------------------------------------------------------------------------------------------------------------------------------------------------------------------


Cell Area Report
-----------------------------------------------------------------------------------------------------------------------------------------------------------------------

Group Names                    All (%)       Repeaters (%)   Combinational (%)       Registers (%)     Sequentials (%)   Clock_network (%)          Macro (%)   Physical_only (%)
-----------------------------------------------------------------------------------------------------------------------------------------------------------------------
saed32cell_hvt          451.87   (8.91%)     152.23   (3.00%)     292.52   (5.77%)      7.12   (0.14%)      0.00   (0.00%)      0.00   (0.00%)      0.00   (0.00%)      0.00   (0.00%)
saed32cell_lvt          634.85  (12.52%)     109.79   (2.17%)     418.32   (8.25%)    106.74   (2.11%)      0.00   (0.00%)      0.00   (0.00%)      0.00   (0.00%)      0.00   (0.00%)  L
saed32cell_svt         3983.20  (78.57%)     120.21   (2.37%)     547.93  (10.81%)    504.22   (9.95%)      0.00   (0.00%)      6.35   (0.13%)      0.00   (0.00%)   2804.48  (55.32%)
-----------------------------------------------------------------------------------------------------------------------------------------------------------------------
Total                  5069.92 (100.00%)     382.23   (7.54%)    1258.78  (24.83%)    618.08  (12.19%)      0.00   (0.00%)      6.35   (0.13%)      0.00   (0.00%)   2804.48  (55.32%)
Low vth groups          634.85  (12.52%)     109.79   (2.17%)     418.32   (8.25%)    106.74   (2.11%)      0.00   (0.00%)      0.00   (0.00%)      0.00   (0.00%)      0.00   (0.00%)
-----------------------------------------------------------------------------------------------------------------------------------------------------------------------
```

**2.40 %**
**LVT (filler 포함)**
249 / 10,391

**28.30 %**
**LVT (logic)**
249 / 880 · area 28.02 %

**Filler**
SHFILL RVT 9,511개가 SVT group에 포함 → 전체 비율 희석.

**Trade-off**
place 24.97 % → 최종 28.30 %, ECO LVT cell 포함 — 허용 28 % 소폭 초과.

# Final PT — Setup

최종 netlist · SPEF로 DMSA 4 scenario 재확인 — setup 위반 path 없음.

report_timing -delay_type max **PASS**

```
pt_shell> report_timing \
?     -delay_type max \
?     -max_paths 20 \
?     -significant_digits 5 \
?     -pba_mode exhaustive
****************************************
Report : timing
        -path_type full
        -delay_type max
        -slack_lesser_than 0.00000
        -max_paths 20
        -pba_mode exhaustive
        -sort_by slack
Design : multi-scenario design
Version: S-2021.06-SP5-2
Date   : Tue Oct 27 16:29:05 2026
****************************************


Start of Master/Worker Task Processing
----------------------------------
Started    : Command execution in scenario 'normal_tt0p78v125c'
Started    : Command execution in scenario 'normal_ff1p16vn40c'
Started    : Command execution in scenario 'normal_tt1p05vn40c'
Started    : Command execution in scenario 'normal_ff0p85v125c'
Succeeded  : Command execution in scenario 'normal_ff1p16vn40c'
Succeeded  : Command execution in scenario 'normal_ff0p85v125c'
Succeeded  : Command execution in scenario 'normal_tt1p05vn40c'
Succeeded  : Command execution in scenario 'normal_tt0p78v125c'
----------------------------------
End of Master/Worker Task Processing

No paths with slack less than 0.00000.
```

**No paths**

**Setup slack < 0**

4 scenario

**4 / 4**

**Scenario succeeded**

ff0p85 · ff1p16 · tt0p78 · tt1p05

# Final PT — Hold

DMSA 4 scenario — hold 위반 path 없음.

report_timing -delay_type min **PASS**

```
pt_shell> report_timing \
?     -delay_type min \
?     -max_paths 20 \
?     -significant_digits 5 \
?     -pba_mode exhaustive
****************************************
Report : timing
        -path_type full
        -delay_type min
        -slack_lesser_than 0.00000
        -max_paths 20
        -pba_mode exhaustive
        -sort_by slack
Design : multi-scenario design
Version: S-2021.06-SP5-2
Date   : Tue Oct 27 16:29:08 2026
****************************************


Start of Master/Worker Task Processing
----------------------------------
Started    : Command execution in scenario 'normal_tt0p78v125c'
Started    : Command execution in scenario 'normal_ff1p16vn40c'
Started    : Command execution in scenario 'normal_tt1p05vn40c'
Started    : Command execution in scenario 'normal_ff0p85v125c'
Succeeded  : Command execution in scenario 'normal_ff1p16vn40c'
Succeeded  : Command execution in scenario 'normal_ff0p85v125c'
Succeeded  : Command execution in scenario 'normal_tt0p78v125c'
Succeeded  : Command execution in scenario 'normal_tt1p05vn40c'
----------------------------------
End of Master/Worker Task Processing

No paths with slack less than 0.00000.
```

## No paths

**Hold slack < 0**

4 scenario

## 4 / 4

**Scenario succeeded**

ff0p85 · ff1p16 · tt0p78 · tt1p05

# Final PT — Constraints

report_constraints -all_violators — 위반 항목 없음.

report_constraints -all_violators · pba exhaustive

PASS

```
pt_shell> report_constraints -all_violators -significant_digits 5 -pba_mode exhaustive

Start of Master/Worker Task Processing
-----------------------------------
Started    : Command execution in scenario 'normal_tt0p78v125c'
Started    : Command execution in scenario 'normal_tt1p05vn40c'
Started    : Command execution in scenario 'normal_ff1p16vn40c'
Started    : Command execution in scenario 'normal_ff0p85v125c'
Succeeded  : Command execution in scenario 'normal_ff0p85v125c'
Succeeded  : Command execution in scenario 'normal_ff1p16vn40c'
Succeeded  : Command execution in scenario 'normal_tt1p05vn40c'
Succeeded  : Command execution in scenario 'normal_tt0p78v125c'
-----------------------------------
End of Master/Worker Task Processing

****************************************
Report : constraint
        -all_violators
        -path slack_only
        -ignore_register_feedback 0.00
        -pba_mode exhaustive
Design : multi_scenario design
Version: S-2021.06-SP5-2
Date   : Tue Oct 27 16:29:09 2026
****************************************

Scenarios: normal_ff0p85v125c normal_ff1p16vn40c normal_tt0p78v125c normal_tt1p05vn40c
```

**0**

**Violators**

4 scenario

**범위**

setup · hold · max transition · max capacitance 등 design rule 전체.

# 09

SECTION

# Challenge

ss scenario까지 켜고 place → CTS → route 도전.

# ss Corner Challenge

ss corner까지 켜고 도전 — timing closure 불가.

report_qor -summary · ON · after route

**VIOLATED**

```
Timing
------------------------------------------------------------------------------
Context                                WNS           TNS           NVE
------------------------------------------------------------------------------
normal_ff0p85v125c (Setup)          0.04655       0.00000             0
normal_ff1p16vn40c (Setup)          0.10862       0.00000             0
normal_ss0p7v125c  (Setup)         -0.85397     -37.32998            90
normal_ss0p95vn40c (Setup)         -0.12160      -0.88580             8
normal_tt0p78v125c (Setup)         -0.02350      -0.15214             9
normal_tt1p05vn40c (Setup)          0.05461       0.00000             0
Design             (Setup)         -0.85397     -37.33832            91

normal_ff0p85v125c (Hold)           0.02640       0.00000             0
normal_ff1p16vn40c (Hold)           0.00273       0.00000             0
normal_ss0p7v125c  (Hold)           0.01831       0.00000             0
normal_ss0p95vn40c (Hold)           0.06866       0.00000             0
normal_tt0p78v125c (Hold)           0.05509       0.00000             0
normal_tt1p05vn40c (Hold)           0.04320       0.00000             0
Design             (Hold)           0.00273       0.00000             0
------------------------------------------------------------------------------

Miscellaneous
------------------------------------------------------------------------------
Cell Area (netlist):                     2481.20787
Cell Area (netlist and physical only):   2481.20800
Nets with DRC Violations:        0
```

| 단계 | Setup WNS (ns) | |
|---|---|---|
| | ss0p7v125c | ss0p95vn40c |
| Place | −1.01706 (110) | −0.11504 (20) |
| CTS | −0.96835 (86) | −0.10709 (8) |
| Route | **−0.85397 (90)** | −0.12160 (9) |

(n) = 위반 path 수 · ON run report_qor -summary 기준

**Challenge**

ss0p7v125c · ss0p95vn40c scenario도 켜고 place → CTS → route까지 도전.

**Result**

ss0p7v125c(0.70 V · 125 °C) WNS −0.85397 ns, Design 위반 path 91개 → ECO로 메울 수 없는 위반폭.

**Limit**

ss corner closure 불가 → 메인 결과는 ss off 기준.

10

SECTION

# Final Results

단계별 QoR, 최종 sign-off 결과와 총정리.

정태웅 · 청주대학교

# Stage-by-Stage QoR

Floorplan → Place → CTS → Route → ECO → Filler 단계별 scenario timing과 area · utilization.

| 단계 | Setup WNS (ns) | | | | Hold WNS (ns) | Cell area (μm²) | Utilization |
|---|---|---|---|---|---|---|---|
| | tt0p78v125c | tt1p05vn40c | ff0p85v125c | ff1p16vn40c | | | |
| **Floorplan** | – | – | – | – | – | 2,230.62 | 0.4383 |
| **Place (place_opt)** | −0.04325 (23) | +0.03151 | +0.00848 | +0.07045 | – | 2,333.04 | 0.4584 |
| **CTS** | +0.00017 | +0.04441 | +0.02505 | +0.09029 | +0.03779 | 2,242.06 | 0.4406 |
| **Route (route_opt)** | −0.00706 (1) | +0.04319 | +0.02268 | +0.08910 | +0.03852 | 2,253.24 | 0.4427 |
| **ECO (ICC2)** | +0.00032 | +0.04314 | +0.02262 | +0.08908 | +0.03854 | 2,263.91 | 0.4448 |
| **Filler (final)** | **+0.00027** | **+0.04312** | **+0.02260** | **+0.08907** | **+0.03854** | 2,265.44 | 0.4451 |

Setup / Hold: 각 단계 report_qor -summary 캡처 기준 (Route 이후 PBA exhaustive), Hold는 Design WNS. Place는 hold 비활성. (n) = 위반 path 수. Place – ECO utilization = cell area / 5,089.23 (core area).

# Final Results

모든 ECO · filler 이후 — 4 scenario timing · DRC · LVS · PG 통과, 최종 PT 위반 없음.

| 항목 | tt0p78v125c | tt1p05vn40c | ff0p85v125c | ff1p16vn40c |
|---|---|---|---|---|
| Setup WNS (ns) | +0.00027 | +0.04312 | +0.02260 | +0.08907 |
| Hold WNS (ns) | +0.13797 | +0.07940 | +0.08163 | +0.03854 |
| Clock latency / skew | 0.11 / 0.11 | 0.06 / 0.06 | 0.06 / 0.06 | 0.04 / 0.04 |
| PT setup · hold · constraint | 위반 없음 (DMSA 4 scenario, PBA exhaustive) | | | |
| check_routes DRC / open | 0 / 0 (932 nets) | | | |
| LVS short / open / floating | 0 / 0 / 0 | | | |
| PG floating · missing via | 0 · 0 / 0 | | | |
| Filler cell | 9,511 (SHFILL*_RVT) | | | |
| Cell area (netlist / +physical) | 2,265.44 / 5,069.92 μm² | | | |
| Utilization | 0.4451 | | | |
| LVT (logic cell) | 28.30 % (249 / 880) — 허용 28 % 소폭 초과, trade-off | | | |

**+0.00027 ns**
**Setup WNS**
tt0p78v125c · PBA

**+0.03854 ns**
**Hold WNS**
ff1p16vn40c

**0 / 0**
**DRC / LVS**
check_routes · check_lvs

**28.30 %**
**LVT (logic)**
허용 28 % · trade-off

# 총정리

spec flow대로 setup부터 최종 PT sign-off까지 완주.

**Flow**

setup → floorplan · PNS → place → CTS → route → StarRC · PT → ICC2 ECO → filler → 최종 PT, place · CTS · route의 문제되는 pre check error 없음.

**Issue ① Half-cycle setup**

route 후 HC_P2N −0.00706 — PT 자동 ECO 0 fixed → U171 sizing · MUX clone · U154 clone · U10 LVT 수동 적용 → +0.00032.

**Issue ② ECO route error**

2차 PT ECO 후 route DRC 10 → 수동 수정으로 0.

**Sign-off**

setup +0.00027 · hold +0.03854 (PBA), DRC · LVS · PG 0, 최종 PT 위반 없음.
Trade-off: LVT 28.30 % · clock latency 0.04 – 0.11 ns.